Malicious Bot Detection on Twitter Using Reinforcement Learning and URL Feature Analysis

A Project Work Submitted to Jawaharlal Nehru Technological University kakinada, kakinada in the Partial fulfillment for the award of the degree of

Bachelor of Technology

In

CSE(IoT&CSBT)

Submitted by

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Certificate

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Declaration

We here by declare that the project report entitled " Malicious Bot Detection on Twitter Using Reinforcement Learning and URL Feature Analysis" under the Guidance of Mrs. K. Anuradha , M.Tech, Assistant Professor Department of CSE(IoT&CSBT) is submitted in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in CSE(IoT&CSBT). This is a record of Bonafide work carried out by us and the results embodied in this project report have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other University or Institute for the award of any other degree or diploma.

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Abstract

Malicious social bots are automated programs that behave like real users on social media platforms and are often used to spread spam, phishing links, fake news, and promotional content. On Twitter, bots commonly use shortened URLs and multiple redirection to hide malicious destinations, making it difficult for normal filters to detect them. Because these bots can imitate human behavior and adjust their posting style, traditional rule-based detection systems are often not sufficient. This project presents a detection framework that combines **reinforcement learning** with **URL feature analysis** to identify malicious bot accounts. Instead of depending only on tweet text or profile details, the system studies URL characteristics such as redirection depth, domain reputation, sharing frequency, and blacklist status. These features provide stronger signals because malicious bots frequently rely on suspicious links to carry out attacks. A learning automata–based reinforcement model is used to observe user actions over time and update their trust score dynamically. The model receives rewards for correct bot/genuine classifications and penalties for wrong decisions, allowing it to improve continuously. A trust computation mechanism using Bayesian learning and evidence combination from neighboring users further strengthens the reliability of detection. The system is implemented using **Java**, **MySQL**, and **Apache Tomcat**, where tweet and URL data are processed, features are extracted, and results are stored in a database. The platform supports automated analysis and report generation, making it suitable for real-time monitoring environments. The modular design also allows easy integration with other analytic components. Overall, the proposed approach increases detection accuracy, reduces false positives, and adapts to changing bot behavior. It provides a scalable and intelligent solution for securing social networks against malicious bot activities and can be extended to multiple platforms and real-time threat monitoring systems in the future.

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List of Abbreviations

RL – Reinforcement Learning

LA – Learning Automata

URL – Uniform Resource Locator

API – Application Programming Interface

DB – Database

RDBMS – Relational Database Management System

JDBC – Java Database Connectivity

UI – User Interface

SQL – Structured Query Language

HTTP – HyperText Transfer Protocol

JSP – Java Server Pages

JVM – Java Virtual Machine

CSV – Comma Separated Values

TPR – True Positive Rate

FPR – False Positive Rate

1. Introduction

1.1 Introduction

Malicious social bots have become a serious threat to online social networks because they can automatically generate posts, share harmful links, and interact with users at a large scale. On Twitter, these bots are often used to spread phishing URLs, fake news, promotional spam, and misleading campaigns. Since they imitate human behavior and posting patterns, it becomes difficult to distinguish them from genuine users using simple rule-based filters, which creates the need for intelligent and adaptive detection mechanisms. Social media platforms like Twitter are widely used for communication and public discussion, but their open nature makes them vulnerable to automated abuse. Bots can operate continuously and influence conversations quickly, causing security, privacy, and trust issues across the network.

Many traditional bot detection systems rely on profile attributes and tweet text analysis. However, attackers can easily manipulate profile details, keywords, hashtags, and posting styles to bypass such filters. Malicious bots also use URL shortening services, re directions, and disguised domains to hide dangerous links inside tweets, encouraging users to click without knowing the true destination. Because of this, text-only and profile-only detection methods are no longer sufficient. In contrast, URL behavior and sharing patterns are harder to fake consistently and provide stronger signals of malicious intent.

Modern detection approaches therefore focus on URL and behavior-based features. URL feature analysis examines properties such as redirection chains, domain reputation, link repetition rate, blacklist status, and spam indicators. These features remain relatively stable compared to surface-level content features. When combined with user activity patterns and interaction signals, they provide a stronger and more reliable basis for identifying suspicious accounts and malicious campaigns.

Reinforcement learning offers an effective and flexible solution because it learns from continuous interaction and feedback. Instead of depending only on fixed labeled datasets,

the model updates its decisions using rewards for correct classifications and penalties for wrong ones. This allows the system to adapt as bot behavior changes over time. When reinforcement learning is combined with URL feature analysis and trust evaluation, it produces a more accurate and dynamic malicious bot detection mechanism.

The project implements this approach using Java, MySQL, and Apache Tomcat to build a practical and scalable detection framework. The system collects tweet and URL data, p reprocesses and extracts key features, computes trust scores, and classifies accounts as bot or genuine through a learning-based model. Results are stored and displayed through a web interface for monitoring and analysis. This integrated design improves detection accuracy, supports real-time usage, and contributes to enhancing security and reliability in social media platforms.

In addition, the system is designed to handle large volumes of social media data efficiently by using a modular processing pipeline. Each stage — data collection, prepossessing, feature extraction, learning, and classification — is separated so that improvements can be added without affecting the entire system. This modular structure makes the framework flexible and easier to maintain. It also allows new features such as sentiment signals, network graph metrics, or temporal posting patterns to be integrated in future upgrades.

Another important aspect of the approach is trust evaluation based on both direct and indirect signals. Direct trust is calculated from the user's own URL sharing behavior and detected risk features, while indirect trust can be derived from neighboring or connected accounts. Combining multiple evidence sources improves decision confidence and reduces false alarms. This is especially useful in social networks where attackers try to appear legitimate by partially behaving like normal users.

The proposed detection model also supports continuous learning, where new data can be periodically fed into the system to refresh feature statistics and retrain the learning component. This helps the detector remain effective even when attackers change strategies, domains, or URL obfuscation techniques. Such adaptability is critical for real-world deployment because bot tactics evolve frequently and static models degrade over time.

The framework is also suitable for integration with cloud environments, where large-scale data storage and distributed processing can be used to improve performance.

Cloud deployment allows the model to process millions of tweets and URLs efficiently and makes the service accessible through web APIs. Such integration is useful for organizations, researchers, and social media monitoring teams who require scalable and remotely accessible detection tools.

From a research perspective, this project opens opportunities to combine reinforcement learning with deep learning and graph analysis methods. Features derived from follower networks, re tweet graphs, and community structures can further strengthen detection accuracy. Hybrid models that merge behavioral, textual, network, and URL features can produce more robust results against advanced bots that try to evade single-feature detectors.

Twitter. The system not only improves detection accuracy but also supports scalability, adaptability, and easy deployment. With further enhancements and real-time data integration, this framework can serve as a strong foundation for next-generation social media security systems.

1.2 Real time Applications

- **Social Media Platform Protection**

The system can be deployed by social media platforms to automatically identify and block malicious bot accounts in real time. This helps reduce spam tweets, fake followers, and automated attacks, thereby improving user trust and platform reliability.

- **Detection of Malicious and Phishing URLs**

By analyzing URL features, the system can detect malicious links shared on Twitter. This protects users from phishing attacks, malware downloads, and fraudulent websites, ensuring safer online interactions.

- **Fake Account and Spam Control**

The reinforcement learning model continuously learns bot behavior patterns, making it effective in detecting newly created fake accounts and spam bots in real time.

- **Cybersecurity Threat Monitoring**

Cybersecurity organizations can integrate this system into their monitoring tools to track suspicious social media activities. It acts as an early warning system for coordinated Cyber attacks and bot-driven campaigns.

- **Election and Misinformation Control**

The system can be used to detect bot-driven misinformation and propaganda campaigns during elections or major public events. This helps reduce the spread of fake news and misleading content.

- **Brand Reputation Management**

Companies and organizations can use this system to monitor social media for bot-based attacks on their brand, such as fake reviews, spam promotions, or malicious links, protecting brand reputation.

2. Literature Review

2.1 Introduction

Several studies have focused on detecting malicious bots on social media platforms such as Twitter by analyzing user behavior, content patterns, and network interactions.

Alhosseini et al. (2024) proposed combining URL analysis with machine learning models to detect phishing and malicious activities on social networks. The results confirmed that URL features significantly enhance detection accuracy, but the approach remained supervised and dependent on labeled data.

Wang et al. (2023) explored Reinforcement Learning techniques for Cybersecurity applications, demonstrating how RL agents can adapt to dynamic threat environments. Their findings showed that RL-based systems reduce false positives over time. However, the study did not specifically focus on social media bot detection or URL feature integration.

Kudugunta and Ferrara (2022) introduced a deep learning-based approach for bot detection using neural networks to analyze temporal and behavioral features. While the model improved detection performance, it required high computational resources and lacked transparency, making real-time deployment challenging for web-based systems.

Ferrara et al. (2021) provided a comprehensive review of social bot detection techniques and discussed their impact on social media ecosystems. The study highlighted the limitations of traditional machine learning models and stressed the need for intelligent systems capable of learning dynamically. However, the work was primarily analytical and did not propose a concrete adaptive implementation.

Thomas et al. (2020) focused on URL-based spam detection by examining malicious links shared on social media platforms. Their research emphasized features such as URL shortening services, redirection paths, and domain reputation. Although effective in identifying malicious URLs, the system did not consider adaptive learning techniques and struggled with zero-day attacks.

Yang et al. (2019) proposed a supervised machine learning framework for detecting spam bots on Twitter by analyzing tweet content, user metadata, and URL characteristics. While their model achieved good accuracy, it required large labeled datasets and frequent retraining, making it less effective against newly emerging bot strategies.

Chu et al. (2018) presented one of the earliest studies on Twitter bot detection by distinguishing human users, bots, and cyborg accounts using behavioral patterns and content-based features. Their work highlighted that automated accounts show distinct posting frequencies and interaction styles. However, the approach relied on static features and lacked adaptability to evolving bot behaviors.

3. SYSTEM STUDY

3.1 Feasibility Study

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.
Three key considerations involved in the feasibility analysis are

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

3.1.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

3.1.2. TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

3.1.3. SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must

not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

3.2 Software and Hardware Requirements

3.2.1 Hardware Requirements

- Processor - Pentium –IV
- RAM - 4 GB (min)
- Hard Disk - 20 GB
- Key Board - Standard Windows Keyboard
- Mouse - Two or Three Button Mouse
- Monitor - SVGA

3.2.2 Software Requirements

- Operating System - Windows XP
- Coding Language - Java/J2EE(JSP,Servlet)
- Front End - J2EE
- Back End - MySQL

3.3 Existing system

- Besel et al. analyzed social botnet attack on Twitter. The authors have presented that social bots use URL shortening services and URL redirection in order to redirect users to malicious web pages. Echeverria and Zhou presented methods to detect, retrieve, and analyze botnet over thousands of users to observe the social behavior of bots. In, a social bot hunter model has been presented based on the user behavioral features, such as follower ratio, the number of URLs, and reputation score. In, a trust model has been designed to detect malicious activities in an OSN. The authors analyzed that

the low trust value of a user indicates that the information spread by the user is considered as untrustworthy.

- In, an MSBD approach has been proposed by considering user behavioral features, such as commenting, liking, and sharing. Madisetty and Desarkar have developed five different convolutional neural network models by considering tweet features. In, a social botnet detection algorithm is proposed by considering spam content in tweets and trust to identify social bots. Gupta et al. designed a framework for detecting spammers in the Twitter network using different machine learning algorithms. In this article, we focus to detect malicious social bots (who perform phishing attacks) by considering various URL-based features using an LA model.

- Several spam-detection approaches have been proposed in the Twitter network to distinguish non spam accounts and spam accounts. Moreover, these studies consider user profile features, which can easily be modified by malicious bots. To avoid feature manipulation, Yang et al. considered social relationships between malicious users and with their neighboring users based on closeness centrality. Moreover, profile features and social interaction features may not help in detecting malicious URLs that are posted by the participants.

- To address the above-mentioned problem, Janabi et al. considered URL-based features (such as URL length, Http-302 status code, and disabling right click) to distinguish legitimate URLs from suspicious URLs. In, a URL-based approach is proposed to detect spam tweets in Twitter based on the tweet content and URL redirection chains. Patil and Patil used decision tree classifiers with statistical features in order to detect malicious URLs. Moreover, social bots may use malicious URL redirections in order to avoid detection.

3.3.1 Disadvantages

- In the existing work, the system considers user profile features, which can easily be modified by malicious bots.
- This system aims to profile features and social interaction features which may not help in detecting malicious URLs that are posted by the participants.

3.4 Proposed System

- ❖ The proposed LA-MSBD algorithm helps to detect malicious social bots accurately (in terms of precision, recall, F-measure, and accuracy) in Twitter. The major contributions are as follows.

- ❖ Analyze the malicious behavior of a participant by considering URL-based features, such as URL redirection, the relative position of URL, frequency of shared URLs, and spam content in URL.

- ❖ Evaluate the trustworthiness of tweets (posted by each participant) by using the Bayesian learning and Dempster–Shafer theory (DST).

- ❖ Design of an LA-MSBD algorithm by integrating a trust model with a set of URL-based features.

- ❖ Performance evaluation of the proposed LA-MSBD algorithm using two Twitter data sets, namely, The Fake Project data set and Social Honeypot data set in terms of precision, recall, F-measure, and accuracy for MSBD in the Twitter networks.

3.4.1 Advantages

- ❖ The malicious behavior of participants is analyzed effectively by considering features extracted from the posted URLs (in the tweets), such as URL redirection, frequency o shared URLs, and spam content in URL, to distinguish between

legitimate and malicious tweets.

- To protect against the malicious social bot attacks, our proposed LA-based malicious social bot detection (LA-MSBD) algorithm integrates a trust computational model with a set of URL-based features for the detection of malicious social bots.

3.5 Software Environment

About Java

Initially the language was called as "oak" but it was renamed as "Java" in 1995. The primary motivation of this language was the need for a platform-independent (i.e., architecture neutral) language that could be used to create software to be embedded in various consumer electronic devices.

- Java is a programmer's language.
- Java is cohesive and consistent.
- Except for those constraints imposed by the Internet environment, Java gives the programmer, full control.

Finally, Java is to Internet programming where C was to system programming.

Importance of Java to the Internet

Java has had a profound effect on the Internet. This is because; Java expands the Universe of objects that can move about freely in Cyberspace. In a network, two categories of objects are transmitted between the Server and the Personal computer. They are: Passive information and Dynamic active programs. The Dynamic, Self-executing programs cause serious problems in the areas of Security and probability. But, Java addresses those concerns and by doing so, has opened the door to an exciting new form of program called the Applet.

Java can be used to create two types of programs

Applications and Applets : An application is a program that runs on our Computer under the operating system of that computer. It is more or less like one creating using C or C++. Java's ability to create Applets makes it important. An Applet is an application designed to be transmitted over the Internet and executed by a Java –compatible web browser. An applet is actually a tiny Java program, dynamically downloaded across the network, just like an image. But the difference is, it is an intelligent program, not just a media file. It can react to the user input and dynamically change.

Features Of Java

Security

Every time you that you download a "normal" program, you are risking a viral infection. Prior to Java, most users did not download executable programs frequently, and those who did scanned them for viruses prior to execution. Most users still worried about the possibility of infecting their systems with a virus. In addition, another type of malicious program exists that must be guarded against. This type of program can gather private information, such as credit card numbers, bank account balances, and passwords. Java answers both these concerns by providing a "firewall" between a network application and your computer. When you use a Java-compatible Web browser, you can safely download Java applets without fear of virus infection or malicious intent.

Portability

For programs to be dynamically downloaded to all the various types of platforms connected to the Internet, some means of generating portable executable code is needed .As you will see, the same mechanism that helps ensure security also helps create portability.

The Byte code

The key that allows the Java to solve the security and portability problems is that the output of Java compiler is Byte code. Byte code is a highly optimized set of instructions designed to be executed by the Java run-time system, which is called the Java Virtual Machine (JVM). That is, in its standard form, the JVM is an interpreter for byte code.

Translating a Java program into byte code helps makes it much easier to run a program in a wide variety of environments. The reason is, once the run-time package exists for a given system, any Java program can run on it.

Although Java was designed for interpretation, there is technically nothing about Java that prevents on-the-fly compilation of byte code into native code. Sun has just completed its Just In Time (JIT) compiler for byte code. When the JIT compiler is a part of JVM, it compiles byte code into executable code in real time, on a piece-by-piece, demand basis. It is not possible to compile an entire Java program into executable code all at once, because Java performs various run-time checks that can be done only at run time. The JIT compiles code, as it is needed, during execution.

Java, Virtual Machine (JVM)

Beyond the language, there is the Java virtual machine. The Java virtual machine is an important element of the Java technology. The virtual machine can be embedded within a web browser or an operating system. Once a piece of Java code is loaded onto a machine, it is verified. As part of the loading process, a class loader is invoked and does byte code verification makes sure that the code that's has been generated by the compiler will not corrupt the machine that it's loaded on. Byte code verification takes place at the end of the compilation process to make sure that is all accurate and correct. So byte code verification is integral to the compiling and executing of Java code.

Overall **Description**

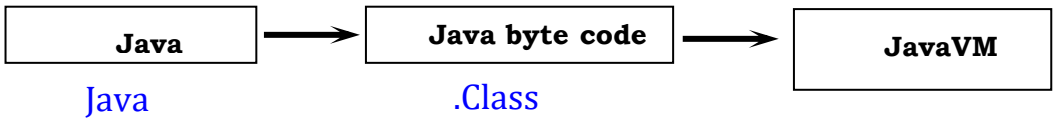


Fig1: JVM

Picture showing the development process of JAVA Program

Java programming uses to produce byte codes and executes them. The first box indicates that the Java source code is located in a. Java file that is processed with a Java compiler called javac. The Java compiler produces a file called a. class file, which contains the byte code. The. Class file is then loaded across the network or loaded locally on your machine into the execution environment is the Java virtual machine, which interprets and executes the byte code.

Java Architecture

Java architecture provides a portable, robust, high performing environment for development. Java provides portability by compiling the byte codes for the Java Virtual Machine, which is then interpreted on each platform by the run-time environment. Java is a dynamic system, able to load code when needed from a machine in the same room or across the planet.

Compilation of code

When you compile the code, the Java compiler creates machine code (called byte code) for a hypothetical machine called Java Virtual Machine (JVM). The JVM is supposed to execute the byte code. The JVM is created for overcoming the issue of portability. The code is written and compiled for one machine and interpreted on all machines. This machine is called Java Virtual Machine.

Compiling and interpreting Java Source Code

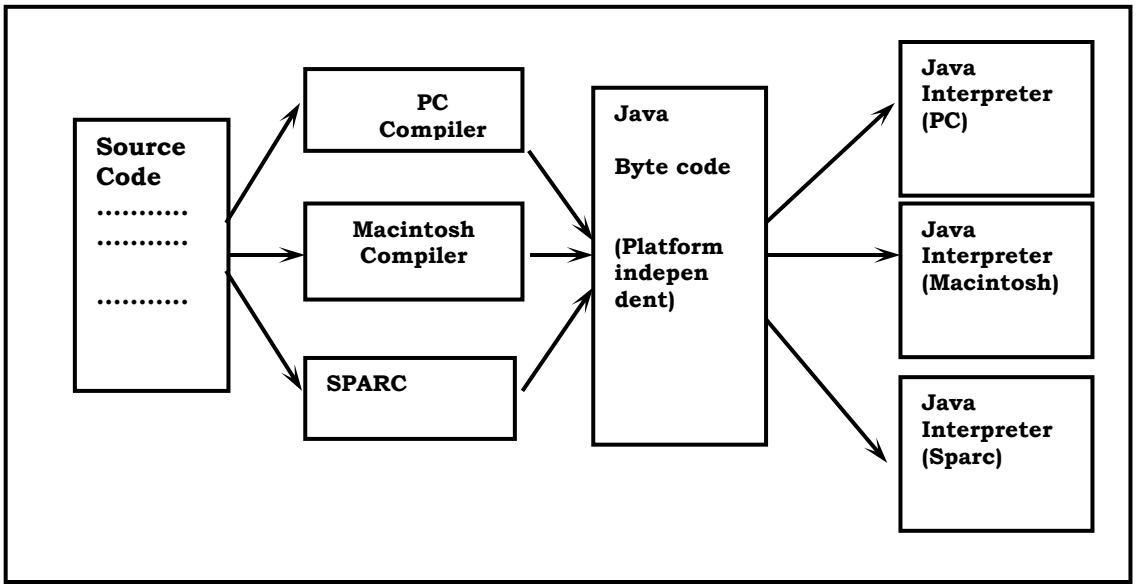


Fig2: Compiling and interpreting Java Source Code

During run-time the Java interpreter tricks the byte code file into thinking that it is running on a Java Virtual Machine. In reality this could be a Intel Pentium Windows 95 or Sun SARC station running Solaris or Apple Macintosh running system and all could receive code from any computer through Internet and run the Applets.

Simple

Java was designed to be easy for the Professional programmer to learn and to use effectively. If you are an experienced C++ programmer, learning Java will be even easier. Because Java inherits the C/C++ syntax and many of the object oriented features of C++. Most of the confusing concepts from C++ are either left out of Java or implemented in a cleaner, more approachable manner. In Java there are a small number of clearly defined ways to accomplish a given task.

Object-Oriented

Java was not designed to be source-code compatible with any other language. This allowed the Java team the freedom to design with a blank slate. One outcome of this was a clean usable, pragmatic approach to objects. The object model in Java is simple and easy to extend, while simple types, such as integers, are kept as high-performance non-objects.

Robust

The multi-platform environment of the Web places extraordinary demands on a program, because the program must execute reliably in a variety of systems. The ability to create robust programs was given a high priority in the design of Java. Java is strictly typed language; it checks your code at compile time and run time.
Java virtually eliminates the problems of memory management and deallocation, which is completely automatic. In a well-written Java program, all run time errors can –and should –be managed by your program.

JAVASCRIPT

JavaScript is a script-based programming language that was developed by Netscape Communication Corporation. JavaScript was originally called Live Script and renamed as JavaScript to indicate its relationship with Java. JavaScript supports the development of both client and server components of Web-based applications. On the client side, it can be used to write programs that are executed by a Web browser within the context of a Web page. On the server side, it can be used to write Web server programs that can process information submitted by a Web browser and then updates the browser's display accordingly

Even though JavaScript supports both client and server Web programming, we prefer JavaScript at Client side programming since most of the browsers supports it. JavaScript is almost as easy to learn as HTML, and JavaScript statements can be included in HTML documents by enclosing the statements between a pair of scripting tags.

<SCRIPTS>..</SCRIPT>.

<SCRIPT LANGUAGE = "JavaScript">

JavaScript statements

</SCRIPT>

Here are a few things we can do with JavaScript :

- Validate the contents of a form and make calculations.
- Add scrolling or changing messages to the Browser's status line.
- Animate images or rotate images that change when we move the mouse over them.
- Detect the browser in use and display different content for different browsers.
- Detect installed plug-ins and notify the user if a plug-in is required.

We can do much more with JavaScript, including creating entire application.

Javascript vs Java

JavaScript and Java are entirely different languages. A few of the most glaring differences are:

- Java applets are generally displayed in a box within the web document; JavaScript can affect any part of the Web document itself.
- While JavaScript is best suited to simple applications and adding interactive features to Web pages; Java can be used for incredibly complex applications.

There are many other differences but the important thing to remember is that

JavaScript and Java are separate languages. They are both useful for different things; in fact they can be used together to combine their advantages.

Advantages

- JavaScript can be used for Sever-side and Client-side scripting.
- It is more flexible than Vb-script.
- JavaScript is the default scripting languages at Client-side since all the browsers supports it.

Java Database Connectivity

What Is JDBC?

JDBC is a Java API for executing SQL statements. (As a point of interest, JDBC is a trademarked name and is not an acronym; nevertheless, JDBC is often thought of as standing for Java Database Connectivity. It consists of a set of classes and interfaces written in the Java programming language. JDBC provides a standard API for tool/database developers and makes it possible to write database applications using a pure Java API.

Using JDBC, it is easy to send SQL statements to virtually any relational database. One can write a single program using the JDBC API, and the program will be able to send SQL statements to the appropriate database. The combinations of Java and JDBC lets a programmer write it once and run it anywhere.

What Does JDBC Do?

Simply put, JDBC makes it possible to do three things:

- Establish a connection with a database
- Send SQL statements
- Process the results.

JDBC versus ODBC and other APIs

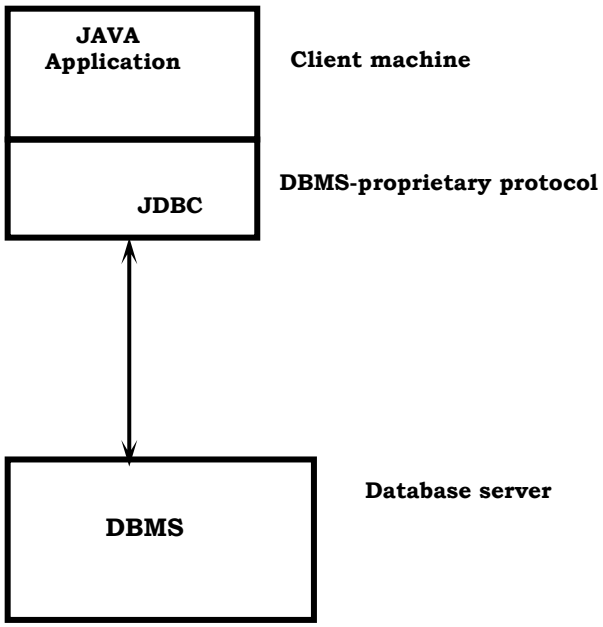
At this point, Microsoft's ODBC (Open Database Connectivity) API is that probably the most widely used programming interface for accessing relational databases. It offers the ability to connect to almost all databases on almost all platforms.

So why not just use ODBC from Java? The answer is that you can use ODBC from Java, but this is best done with the help of JDBC in the form of the JDBC-ODBC Bridge, which we will cover shortly. The question now becomes "Why do you need JDBC?" There are several answers to this question:

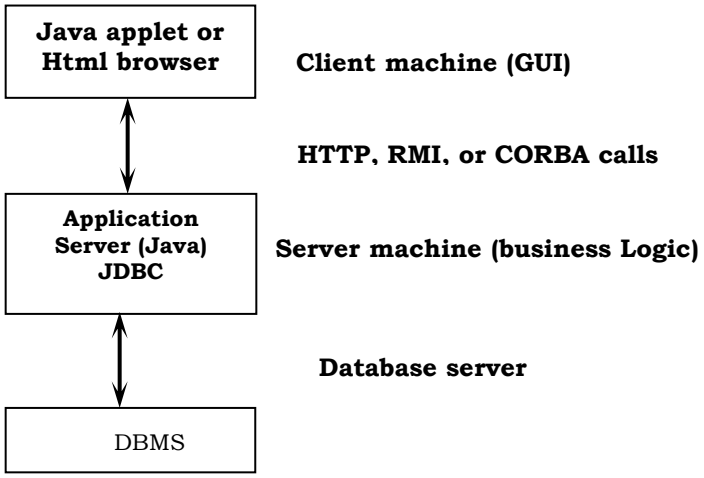
1. ODBC is not appropriate for direct use from Java because it uses a C interface. Calls from Java to native C code have a number of drawbacks in the security, implementation, robustness, and automatic portability of applications.
2. A literal translation of the ODBC C API into a Java API would not be desirable. For example, Java has no pointers, and ODBC makes copious use of them, including the notoriously error-prone generic pointer "void *". You can think of JDBC as ODBC translated into an object-oriented interface that is natural for Java programmers.
3. ODBC is hard to learn. It mixes simple and advanced features together, and it has complex options even for simple queries. JDBC, on the other hand, was designed to keep simple things simple while allowing more advanced capabilities where required.
4. A Java API like JDBC is needed in order to enable a "pure Java" solution. When ODBC is used, the ODBC driver manager and drivers must be manually installed on every client machine. When the JDBC driver is written completely in Java, however, JDBC code is automatically installable, portable, and secure on all Java platforms from network computers to mainframes.

Two-tier and Three-tier Models

The JDBC API supports both two-tier and three-tier models for database access. In the two-tier model, a Java applet or application talks directly to the database. This requires a JDBC driver that can communicate with the particular database management system being accessed. A user's SQL statements are delivered to the database, and the results of those statements are sent back to the user. The database may be located on another machine to which the user is connected via a network. This is referred to as a client/server configuration, with the user's machine as the client, and the machine housing the database as the server. The network can be an Intranet, which, for example, connects employees within a corporation, or it can be the Internet.



In the three-tier model, commands are sent to a "middle tier" of services, which then send SQL statements to the database. The database processes the SQL statements and sends the results back to the middle tier, which then sends them to the user. MIS directors find the three-tier model very attractive because the middle tier makes it



possible to maintain control over access and the kinds of updates that can be made to corporate data. Another advantage is that when there is a middle tier, the user can employ an easy-to-use higher-level API which is translated by the middle tier into the appropriate low-level calls. Finally, in many cases the three-tier architecture can provide performance advantages.

Until now the middle tier has typically been written in languages such as C or C++,

which offer fast performance. However, with the introduction of optimizing compilers that translate Java byte code into efficient machine-specific code, it is becoming practical to implement the middle tier in Java. This is a big plus, making it possible to take advantage of Java's robustness, multi-threading, and security features. JDBC is important to allow database access from a Java middle tier.

JDBC Driver Types

The JDBC drivers that we are aware of at this time fit into one of four categories:

- JDBC-ODBC bridge plus ODBC driver
- Native-API partly-Java driver
- JDBC-Net pure Java driver
- Native-protocol pure Java driver

JDBC-ODBC Bridge

If possible, use a Pure Java JDBC driver instead of the Bridge and an ODBC driver. This completely eliminates the client configuration required by ODBC. It also eliminates the potential that the Java VM could be corrupted by an error in the native code brought in by the Bridge (that is, the Bridge native library, the ODBC driver manager library, the ODBC driver library, and the database client library).

What Is the JDBC- ODBC Bridge?

The JDBC - ODBC Bridge is a JDBC driver, which implements JDBC operations by translating them into ODBC operations. To ODBC it appears as a normal application program. The Bridge implements JDBC for any database for which an ODBC driver is available.

sun.jdbc.odbc Java package and contains a native library used to access ODBC. The Bridge is a joint development of Intersolv and Java Soft.

Java Server Pages (JSP)

Java server Pages is a simple, yet powerful technology for creating and maintaining dynamic-content web pages. Based on the Java programming language, Java Server Pages offers proven portability, open standards, and a mature re-usable component model .The Java Server Pages architecture enables the separation of content

generation from content presentation. This separation not eases maintenance headaches, it also allows web team members to focus on their areas of expertise. Now, web page designer can concentrate on layout, and web application designers on programming, with minimal concern about impacting each other's work.

Features of JSP

Portability:

Java Server Pages files can be run on any web server or web-enabled application server that provides support for them. Dubbed the JSP engine, this support involves recognition, translation, and management of the Java Server Page lifecycle and its interaction components.

Components

It was mentioned earlier that the Java Server Pages architecture can include reusable Java components. The architecture also allows for the embedding of a scripting language directly into the Java Server Pages file. The components current supported include Java Beans, and Servlets.

Processing

A Java Server Pages file is essentially an HTML document with JSP scripting or tags. The Java Server Pages file has a JSP extension to the server as a Java Server Pages file. Before the page is served, the Java Server Pages syntax is parsed and processed into a Servlet on the server side. The Servlet that is generated outputs real content in straight HTML for responding to the client.

Access Models:

A Java Server Pages file may be accessed in at least two different ways. A client's request comes directly into a Java Server Page. In this scenario, suppose the page accesses reusable Java Bean components that perform particular well-defined computations like accessing a database. The result of the Beans computations, called result sets is stored within the Bean as properties. The page uses such Beans to generate dynamic content and present it back to the client.

In both of the above cases, the page could also contain any valid Java code. Java Server Pages architecture encourages separation of content from presentation.

Steps in the execution of a JSP Application:

- The client sends a request to the web server for a JSP file by giving the name of the JSP file within the form tag of a HTML page.
- This request is transferred to the Java Web Server. At the server side Java Web Server receives the request and if it is a request for a jsp file server gives this request to the JSP engine.
- JSP engine is program which can understands the tags of the jsp and then it converts those tags into a servlet program and it is stored at the server side. This servlet is loaded in the memory and then it is executed and the result is given back to the JavaWebServer and then it is transferred back to the result is given back to the JavaWebServer and then it is transferred back to the client.

JDBC connectivity

The JDBC provides database-independent connectivity between the J2EE platform and a wide range of tabular data sources. JDBC technology allows an Application Component Provider to:

- Perform connection and authentication to a database server
- Manager transactions
- Move SQL statements to a database engine for preprocessing and execution
- Execute stored procedures
- Inspect and modify the results from Select statements.

4 . MODULE / METHODOLOGY

4.1 Modules

4.1.1 Tweet User

In this module, the Admin has to login by using valid user name and password. After login successful he can perform some operations such as view and authorize users, Adding Short URLs, Listing all Friends Request and Responses, Listing all User Posted Tweets, Listing all Tweets and Re-tweets with Comments ,Viewing all Spammers URLs, Viewing URL Shortening Users and Post Details, Finding all Clicked Shortened URLs and Corresponding Users and Chart Results.

Viewing and Authorizing Users

In this module, the admin views all users details and authorize them for login permission. User Details such as User Name, Address, Email Id, Mobile Number.

Viewing all Friends Request and Response

In this module, the admin can see all the friends' requests and response history. Details such as Requested User Name and Image, and Requested to User Name and Image, status and date.

List all User Posted Tweets

In this module, the admin can see all the tweets posted by the users. The Tweet Details such as, tweet name, tweet image, tweet description, tweet uses, date of post and Posted user name.

View all Inference Attackers

In this module, the all Inference Attacker details will be listed. The details consist of the comment which has Shortening URLs (like t.co, goo.gl, bit.ly), Tweet Name, and Date of Attack.

View URL Shortening Users and Post Details

In this, the admin can see all URL Shortening users and post details. This contains the number of times the particular user used these URLs (t.co, goo.gl, bit.ly) while commenting on tweets.

View all Clicked Shortened URLs and Corresponding End Users

In this, the admin can view all the users who clicked Number of times on these URLs (t.co, goo.gl, bit.ly).

Find Number of times Posted URL shortening users in Chart

In this, the admin can see the graph which describes the number of times the particular user used these Shortened URLs while tweeting or Re-Tweeting (Posting their comment).

Find Number of times used URL shortening users in Chart

In this, the admin can see the graph which describes the number of times the particular Shortened URL is used by the users while tweeting or Re-Tweeting (Posting their comment).

4.1.2 User

In this module, there are n numbers of users are present. User should register before performing any operations. Once user registers, their details will be stored to the database. After registration successful, he has to login by using authorized user name and password. Once Login is successful user can perform some operations like viewing their profile details, searching for friends and sending friend requests, accepting friend requests, viewing friends details, Posting Their own Tweets, Finding Friends tweets and Re-tweets, Listing user tweets and comments and Finding Inference Attack user Posted tweets.

Viewing Profile Details, Search and Request Friends

In this module, the user can see their own profile details, such as their address, email, mobile number, profile Image.

The user can search for friends and can send friend requests or can accept friend

requests.

Post Tweets

In this, the user can post their own tweets by providing details such as tweet image, tweet name, tweet description, tweet uses.

View Friends Tweets on Posts and Re-Tweet

In this, the user can see all his/her friends' tweets on posts and can Re-tweet on them by providing user own comment (if the comment contains Shortened URLs that is, t.co, goo.gl, bit.ly then user will become an inference attacker).

View Inference Attack on User Posts(Tweets)

In this, the user can see all the Inference attackers who have posted Shortening URLs in their comments on User Posts.

5 . SYSTEM DESIGN

5.1 Architecture Diagram

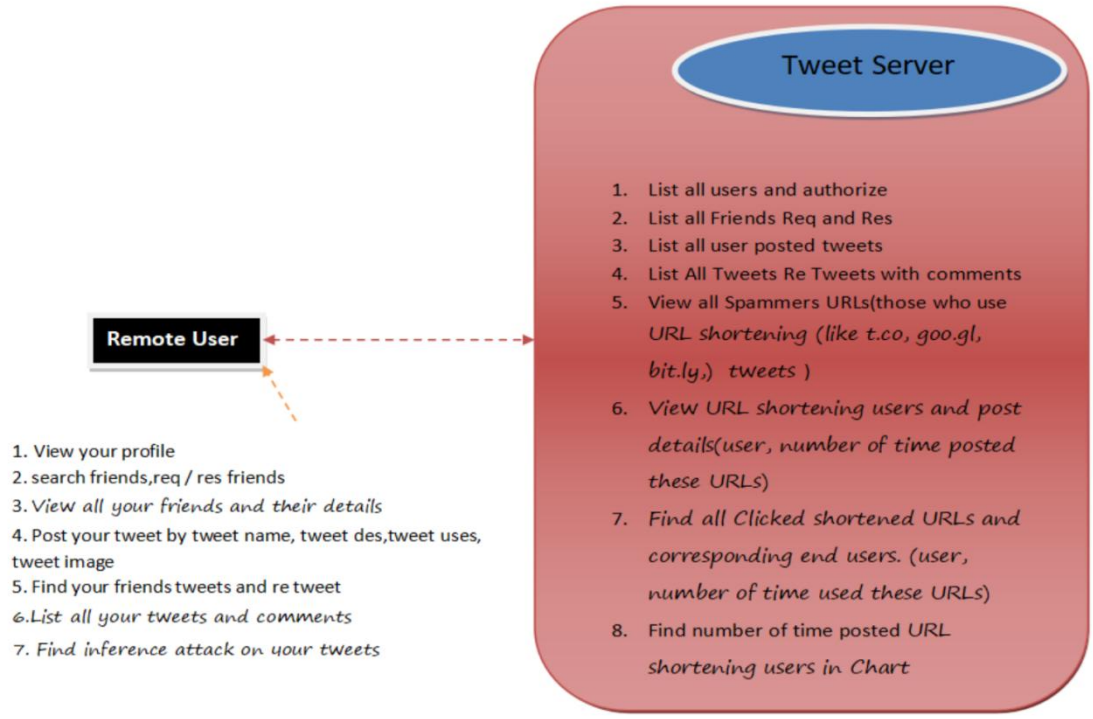


Fig3: Architecture Design

5.2 UML Diagrams

5.2.1 Usecase Diagram

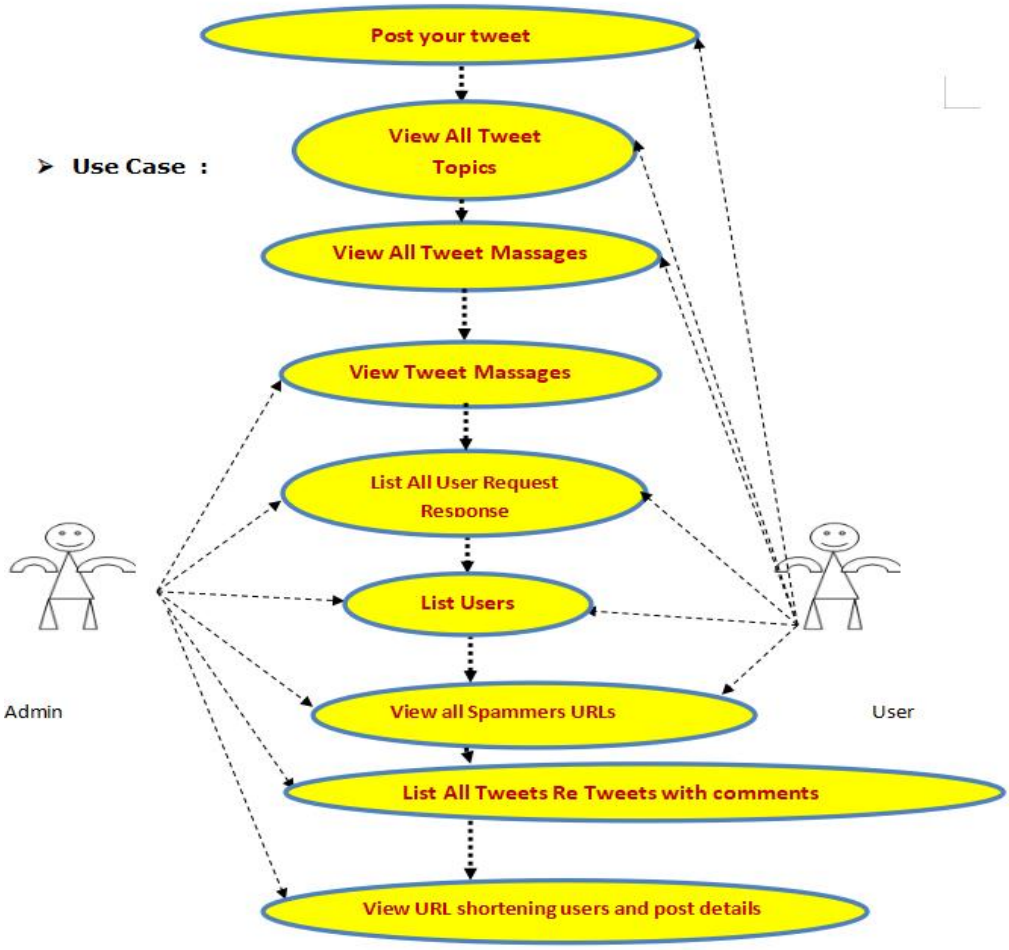


Fig4: Usecase Diagram

5.2.2 Class Diagram

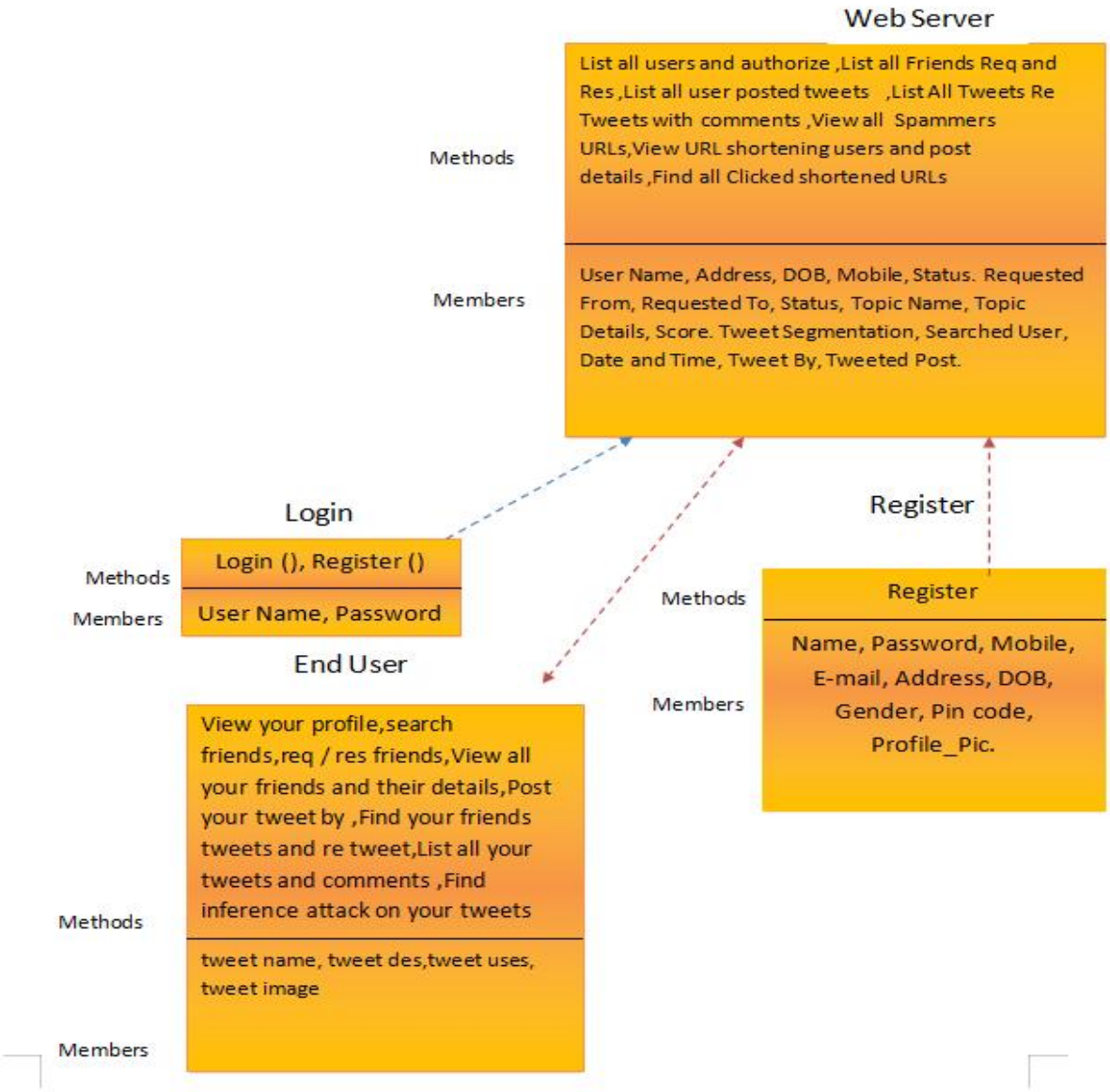


Fig4 : Class Diagram

5.2.3 Sequence Diagram

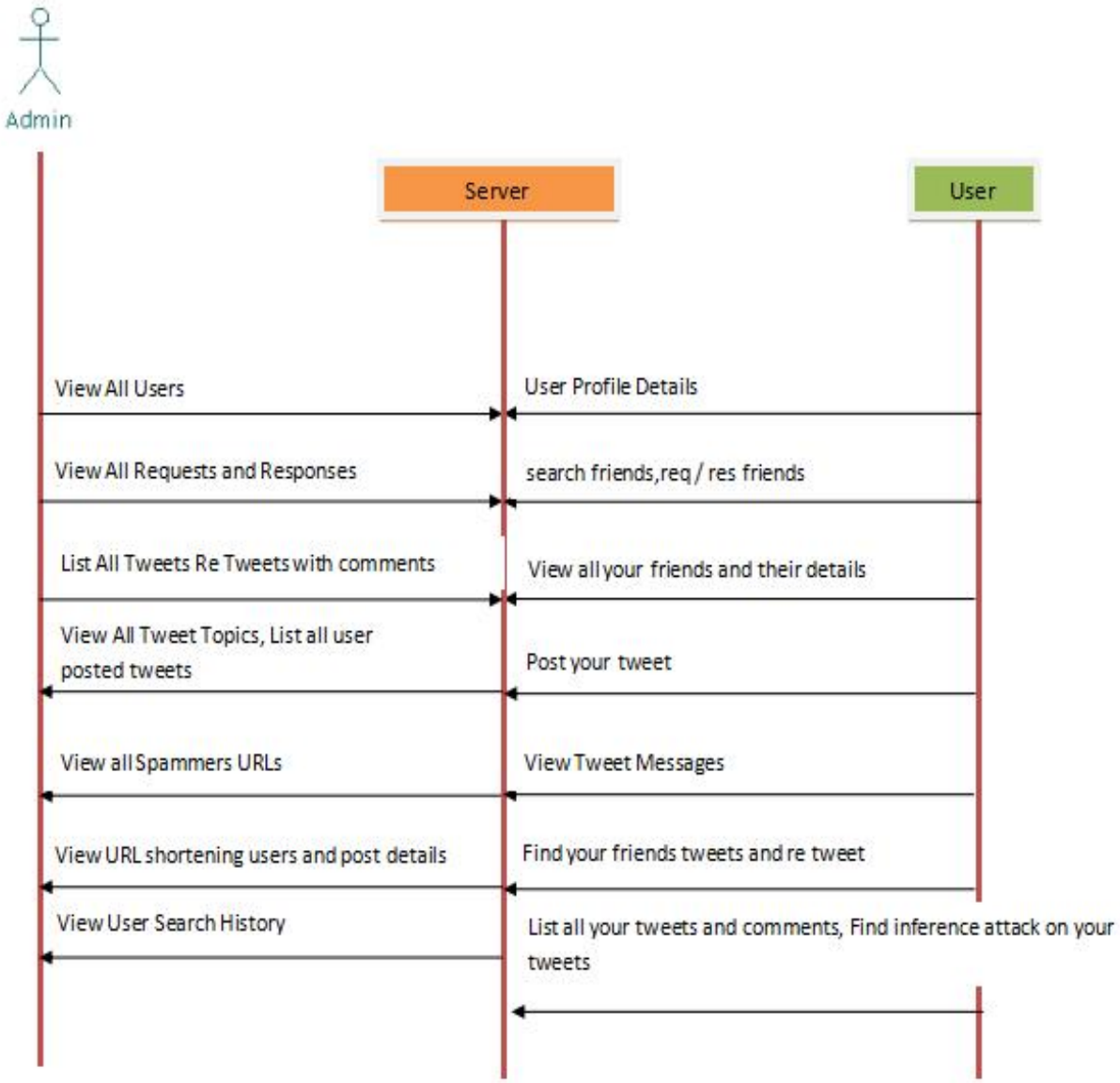


Fig5 : Sequence Diagram

5.3 Flow chart Diaram

5.3.1 User Flowchart

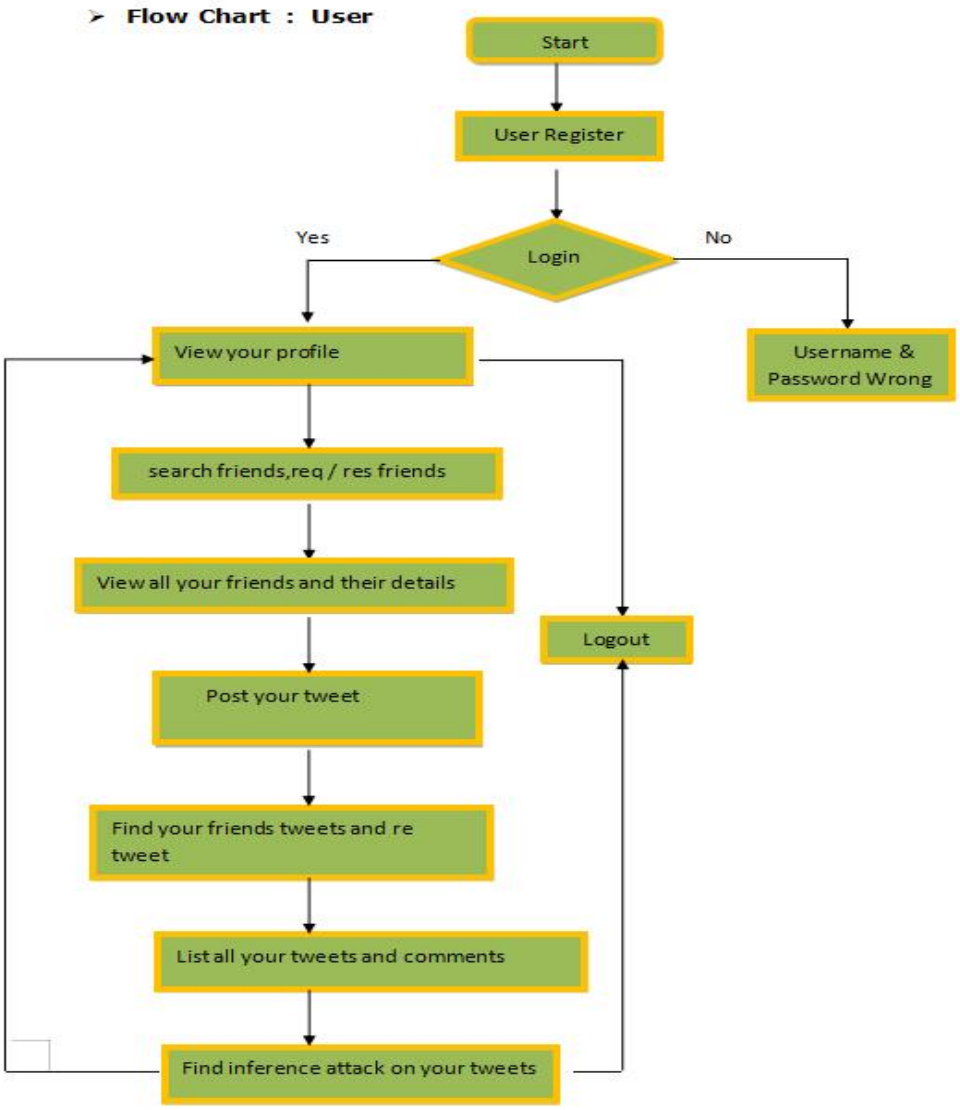


Fig6: UserFlow chart

5.3.2 Admin Flowchart

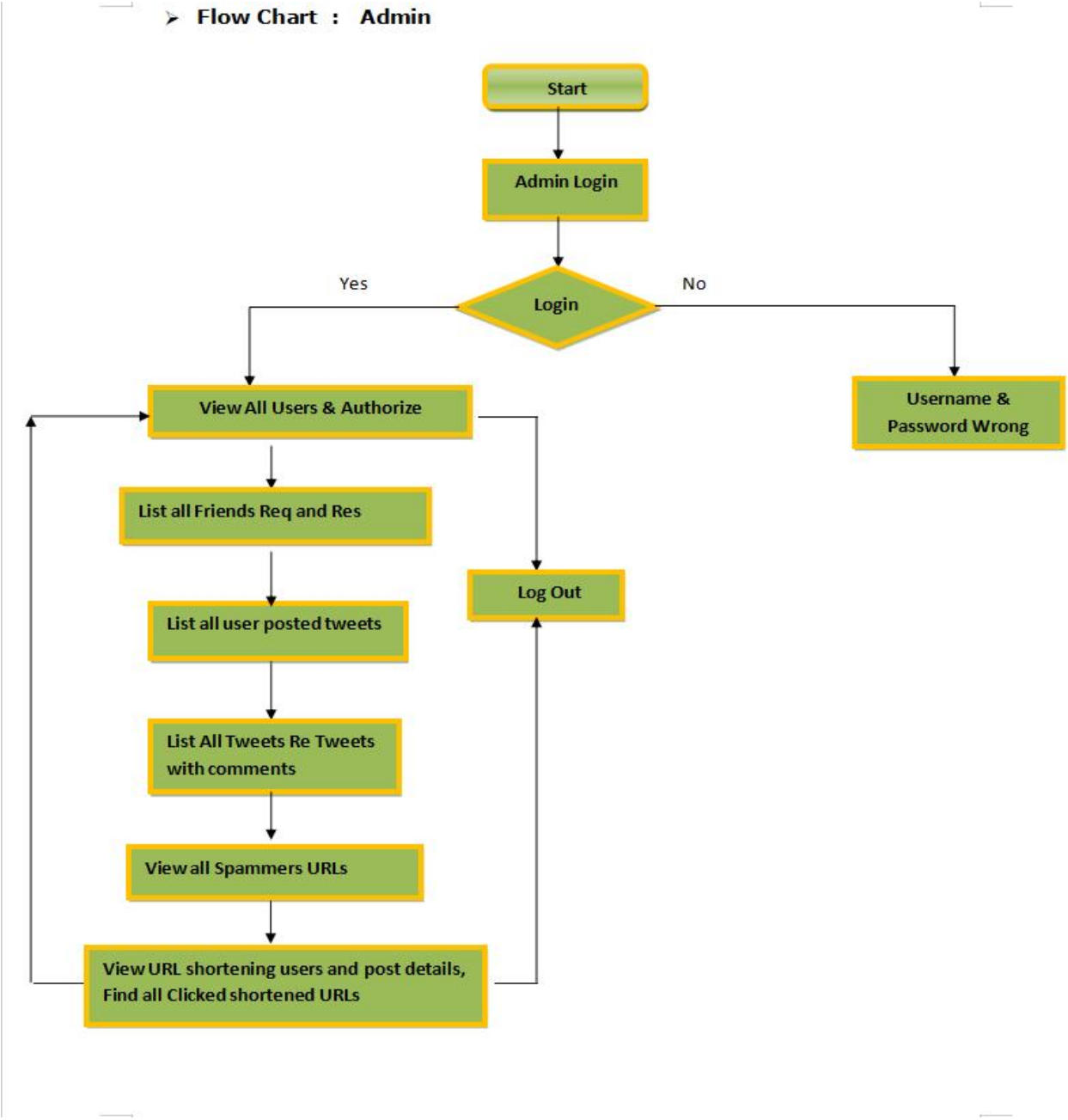


Fig7: AdminFlow chart

5.4 Data Flow Diagram

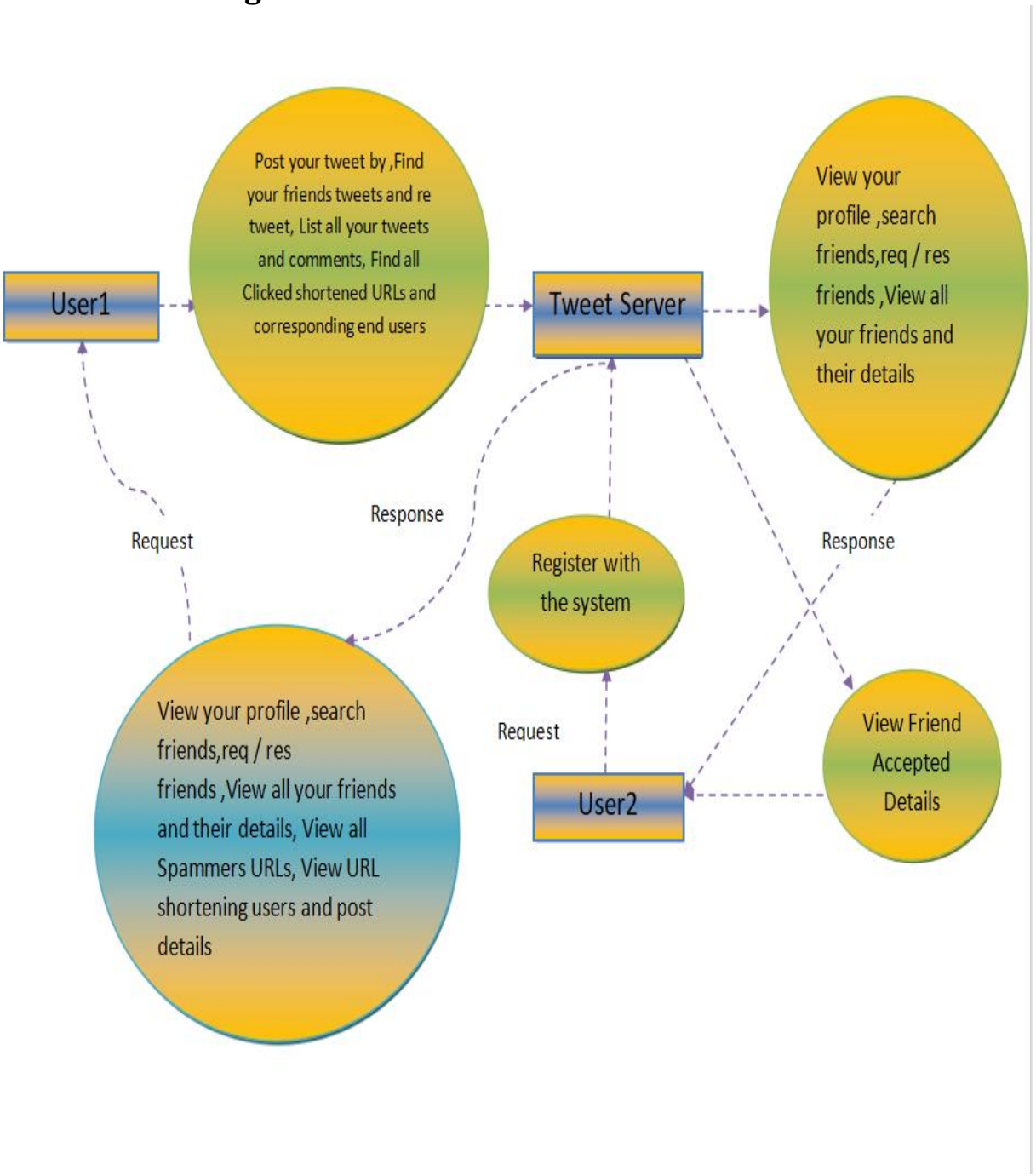


Fig9: Data Flow Diagram

6. Implementation / Source code

 # DB Connection

```
<%@page import="java.sql.*"%>
<%
Connection con=null;
try{
 Class.forName("com.mysql.jdbc.Driver");
 con=DriverManager.getConnection(
  "jdbc:mysql://localhost:3306/botdb","root","root");
}catch(Exception e){ out.println(e); }
%>
```

 # Registration

```
<form action="register.jsp" method="post" enctype="multipart/form-data">
<input name="u" placeholder="User" required>
<input type="password" name="p" placeholder="Pass" required>
<input name="e" placeholder="Email">
<button>Register</button>
</form>

<%@include file="DB.jsp"%>
<%
PreparedStatement ps=con.prepareStatement(
"insert into user(username,password,email) values(?,?,?)");
ps.setString(1,request.getParameter("u"));
ps.setString(2,request.getParameter("p"));
ps.setString(3,request.getParameter("e"));
ps.executeUpdate();
%>
```

 # Login

```
<form method="post">
<input name="u">
<input type="password" name="p">
<button>Login</button>
</form>

<%@include file="DB.jsp"%>
<%
PreparedStatement ps=con.prepareStatement(
"select * from user where username=? and password=?");
ps.setString(1,request.getParameter("u"));
ps.setString(2,request.getParameter("p"));
ResultSet rs=ps.executeQuery();
if(rs.next()){
 session.setAttribute("uname",rs.getString(1));
 response.sendRedirect("home.jsp");
```

```
}

%>
```

 # Search Friends

```
<%@include file="DB.jsp"%>
<%
PreparedStatement ps=con.prepareStatement(
"select username from user where username like ?");
ps.setString(1,"%"+request.getParameter("k")+"%");
ResultSet rs=ps.executeQuery();
while(rs.next()) out.println(rs.getString(1)+"<br>");
%>
```

#Post Tweet

```
<%@include file="DB.jsp"%>
<%
String t=request.getParameter("tweet");
PreparedStatement ps=con.prepareStatement(
"insert into tweet(user,text) values(?,?)");
ps.setString(1,(String)session.getAttribute("uname"));
ps.setString(2,t);
ps.executeUpdate();
%>
```

 # URL Detection

```
<%
String txt=request.getParameter("tweet");
if(txt.matches(".*(bit.ly|t.co|goo.gl).*")){
 out.println("Short URL detected");
}
%>
```

7 . TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

7.1 TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional testing

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input	:	identified classes of valid input must be accepted.
Invalid Input	:	identified classes of invalid input must be rejected.
Functions	:	identified functions must be exercised.
Output	:	identified classes of application outputs must be exercised.
Systems/Procedures	:	interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Testing

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner

workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot "see" into it. The test provides inputs and responds to outputs without considering how the software works.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Output Testing

After performing the validation testing, the next step is output testing of the proposed system, since no system could be useful if it does not produce the required output in the specified format. Asking the users about the format required by them tests the outputs generated or displayed by the system under consideration. Hence the output format is considered in 2 ways – one is on screen and another in printed format.

Validation Checking

Validation checks are performed on the following fields.

Text Field

The text field can contain only the number of characters lesser than or equal to its size. The text fields are alphanumeric in some tables and alphabetic in other tables. Incorrect

entry always flashes and error message.

Numeric Field

The numeric field can contain only numbers from 0 to 9. An entry of any character flashes an error messages. The individual modules are checked for accuracy and what it has to perform. Each module is subjected to test run along with sample data. The individually tested modules are integrated into a single system. Testing involves executing the real data information is used in the program the existence of any program defect is inferred from the output. The testing should be planned so that all the requirements are individually tested.

A successful test is one that gives out the defects for the inappropriate data and produces and output revealing the errors in the system.

7.2 Preparation of Test Data

Taking various kinds of test data does the above testing. Preparation of test data plays a vital role in the system testing. After preparing the test data the system under study is tested using that test data. While testing the system by using test data errors are again uncovered and corrected by using above testing steps and corrections are also noted for future use.

7.2.1 Using Live Test Data

Live test data are those that are actually extracted from organization files. After a system is partially constructed, programmers or analysts often ask users to key in a set of data from their normal activities. Then, the systems person uses this data as a way to partially test the system. In other instances, programmers or analysts extract a set of live data from the files and have them entered themselves.

It is difficult to obtain live data in sufficient amounts to conduct extensive testing. And, although it is realistic data that will show how the system will perform for the typical processing requirement, assuming that the live data entered are in fact typical, such data generally will not test all combinations or formats that can enter the system. This bias toward typical values then does not provide a true systems test and in fact ignores the cases most likely to cause system failure.

7.2.2 Using Artificial Test Data

Artificial test data are created solely for test purposes, since they can be generated to test all combinations of formats and values. In other words, the artificial data, which can quickly be prepared by a data generating utility program in the information systems department, make possible the testing of all login and control paths through the program. The most effective test programs use artificial test data generated by persons other than those who wrote the programs. Often, an independent team of testers formulates a testing plan, using the systems specifications. The package "Virtual Private Network" has satisfied all the requirements specified as per software requirement specification and was accepted.

7.2.3 User training

Whenever a new system is developed, user training is required to educate them about the working of the system so that it can be put to efficient use by those for whom the system has been primarily designed. For this purpose the normal working of the project was demonstrated to the prospective users. Its working is easily understandable and since the expected users are people who have good knowledge of computers, the use of this system is very easy.

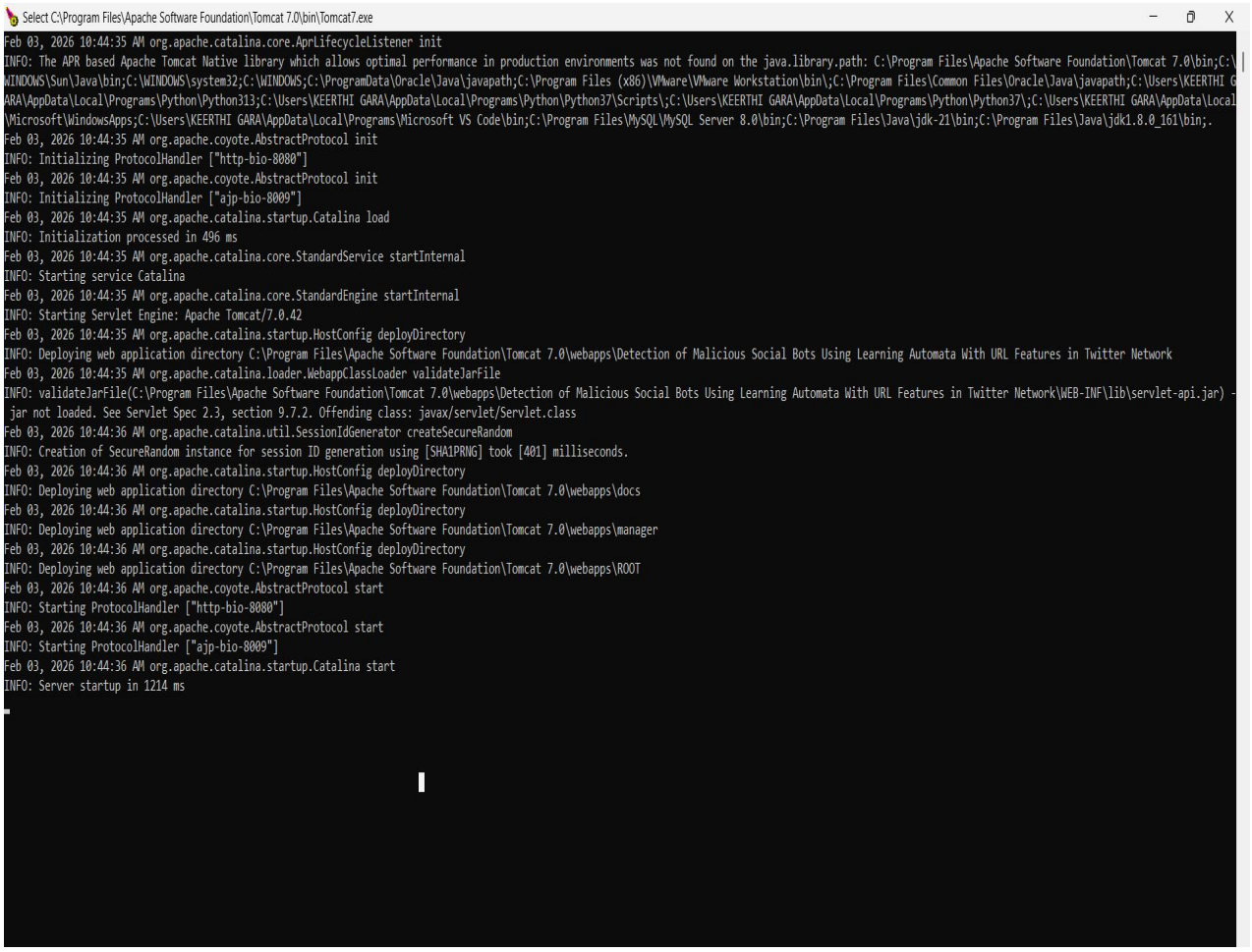
7.3 Testing strategy

A strategy for system testing integrates system test cases and design techniques into a well planned series of steps that results in the successful construction of software. The testing strategy must co-operate test planning, test case design, test execution, and the resultant data collection and evaluation .A strategy for software testing must accommodate low-level tests that are necessary to verify that a small source code segment has been correctly implemented as well as high level tests that validate major system functions against user requirements.

Software testing is a critical element of software quality assurance and represents the ultimate review of specification design and coding. Testing represents an interesting anomaly for the software. Thus, a series of testing are performed for the proposed system before the system is ready for user acceptance testing.

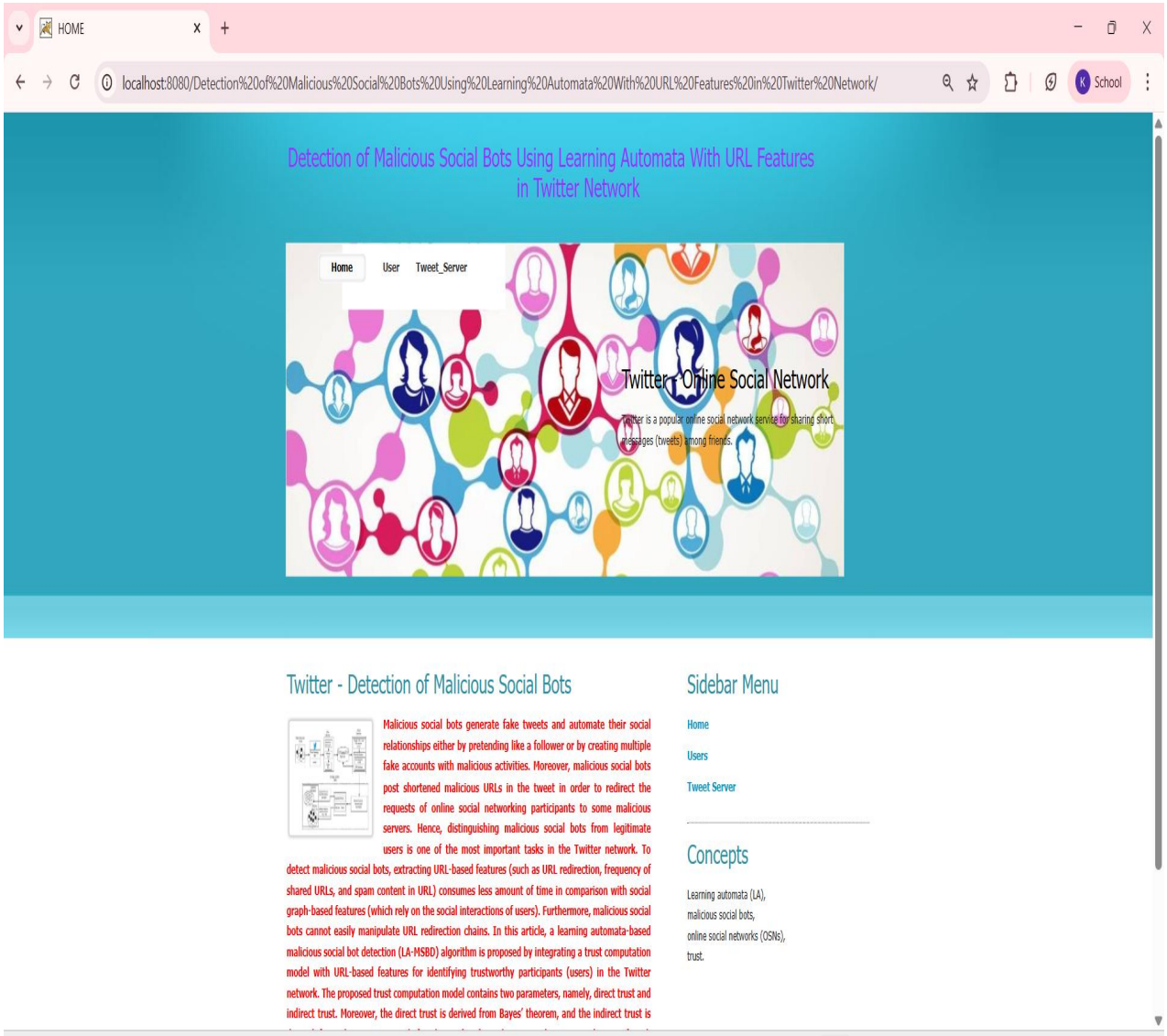
8. RESULTS AND SCREENSHOTS

Tomcat Page



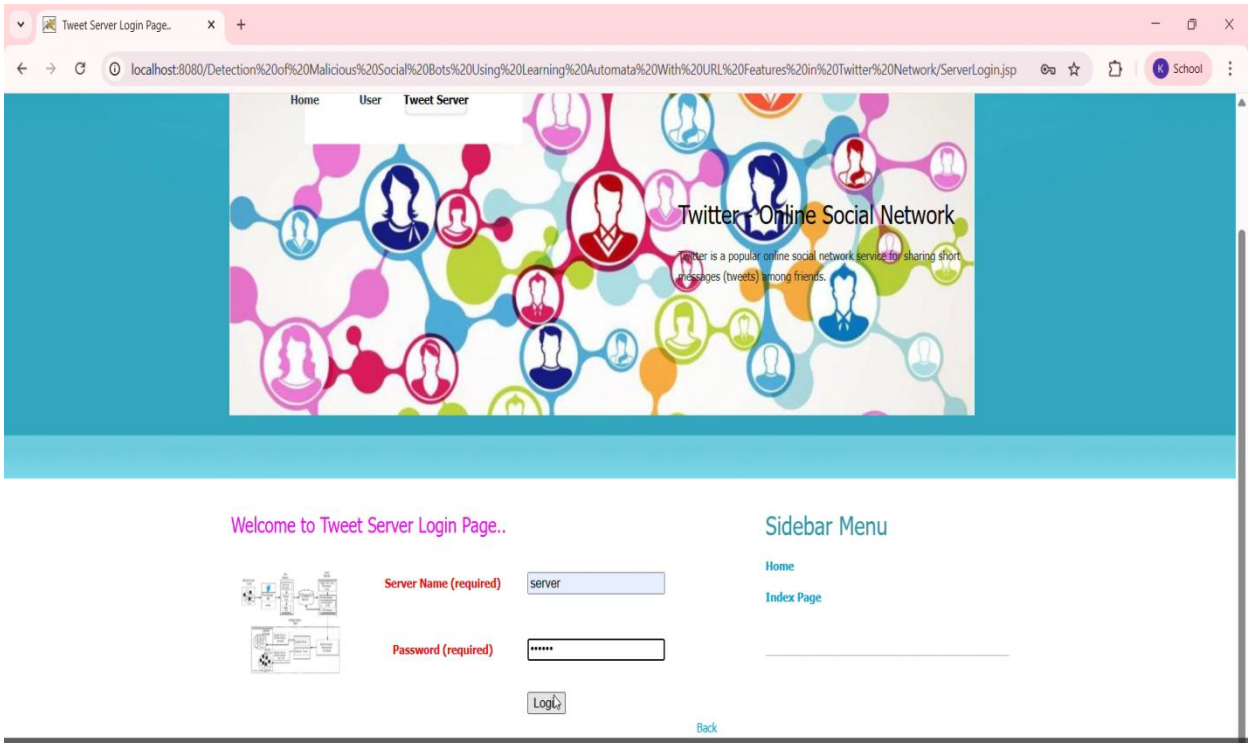
Tomcat

Local Host Page



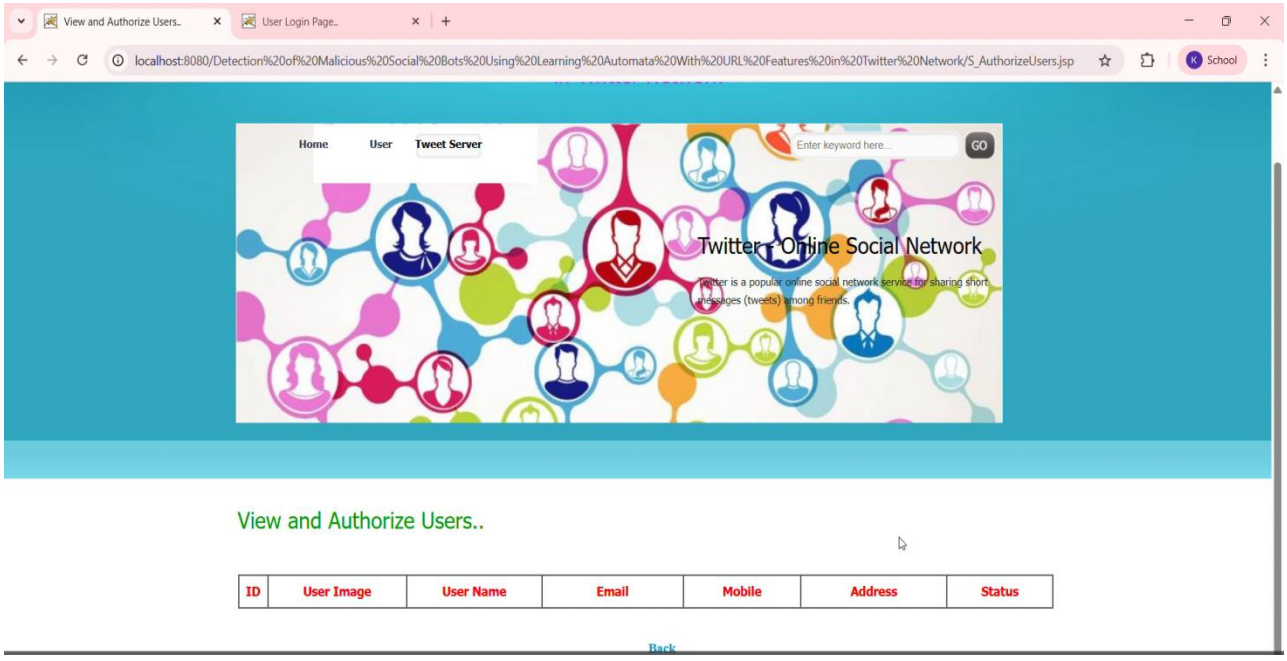
Local Host

Server Login Page



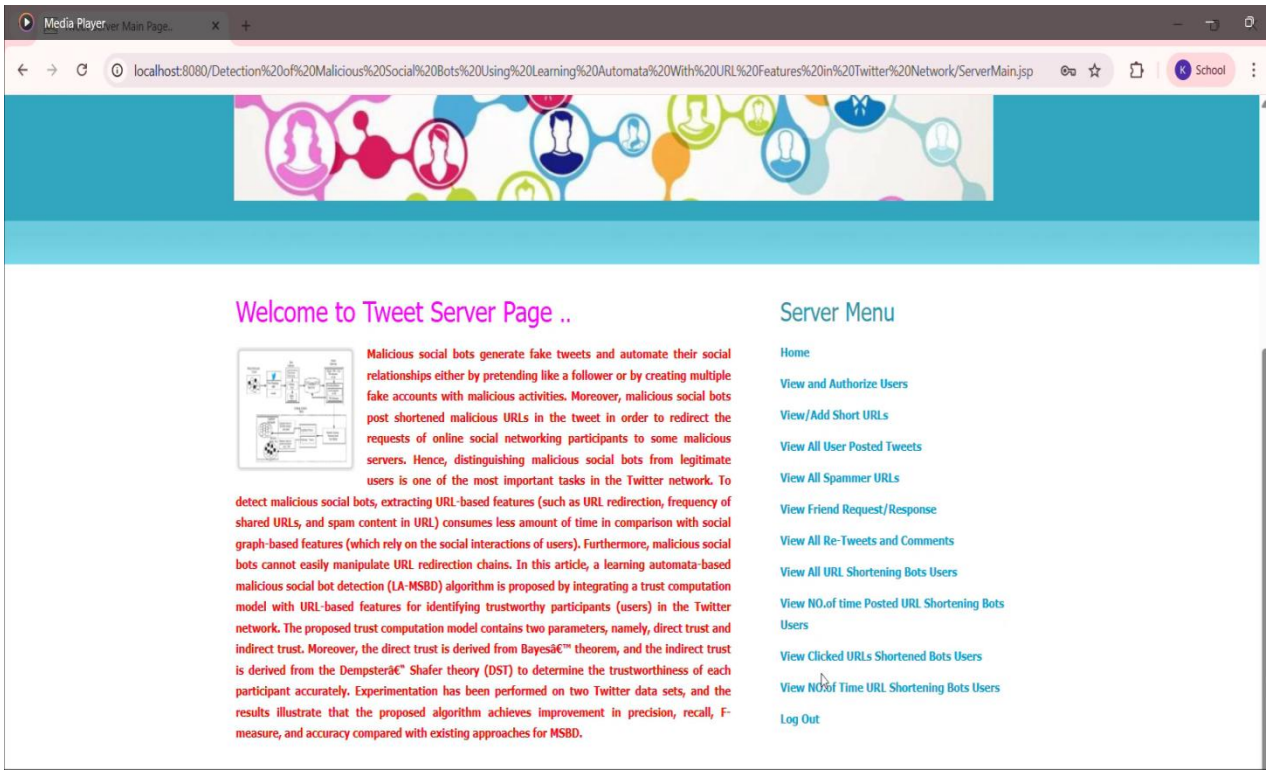
Server Login

View & Author Users Page



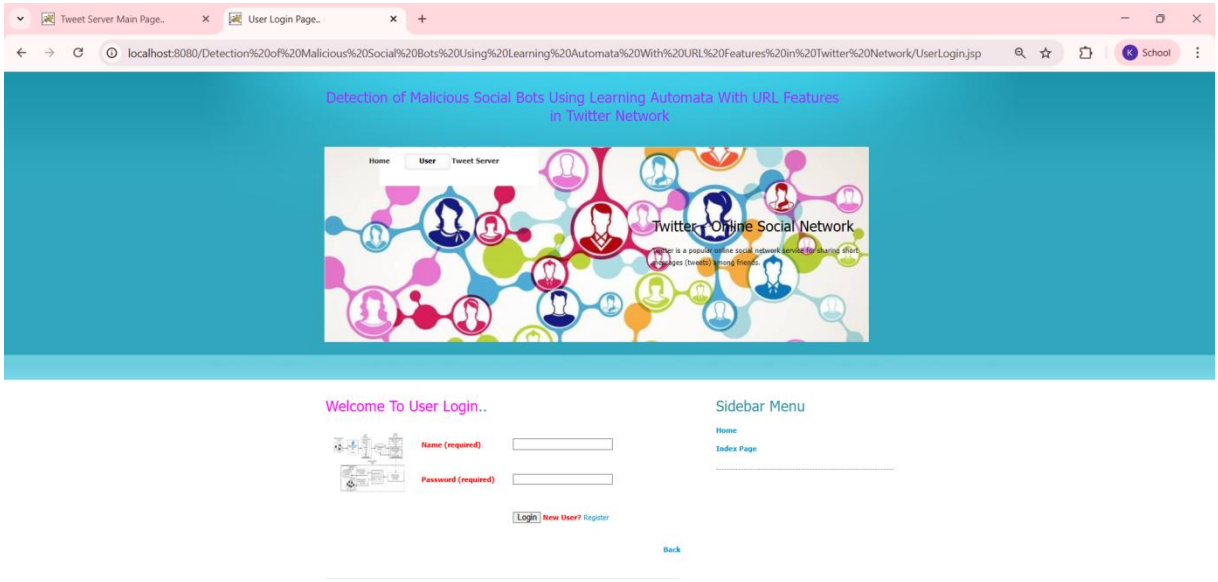
View & Author Users

Server Menu



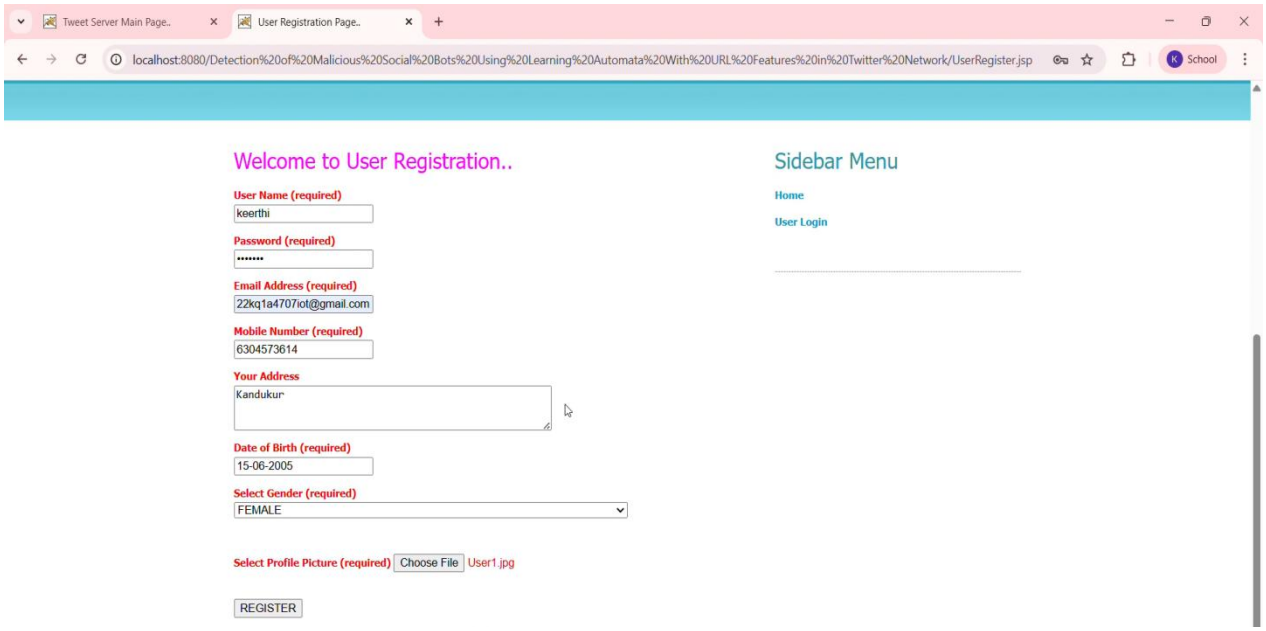
Serevr Menu

User Login & Registration Page



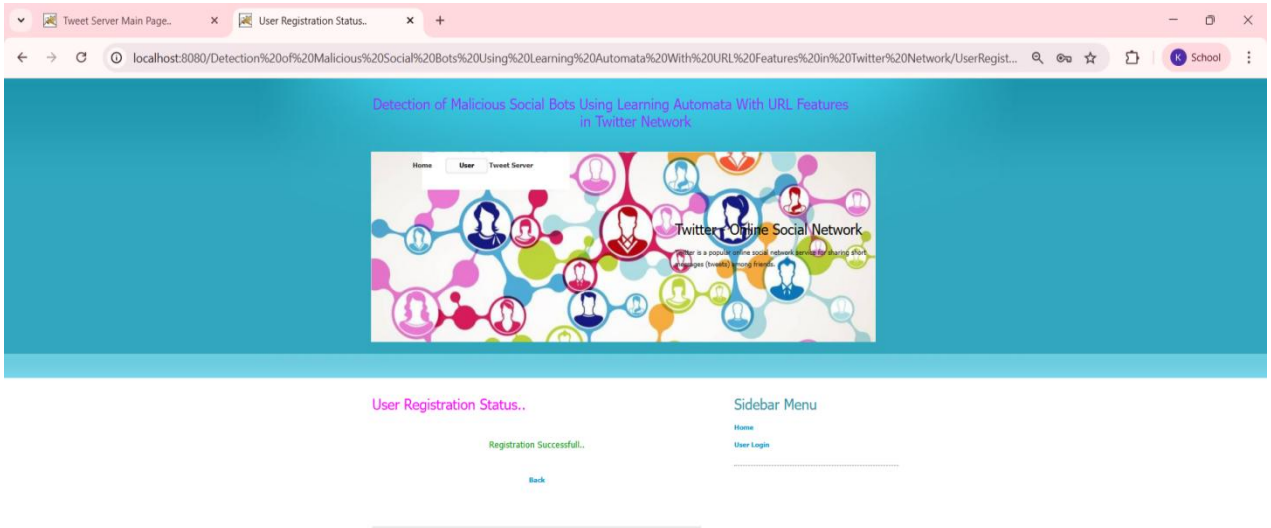
User Login & Registration Page

User Regitration Page



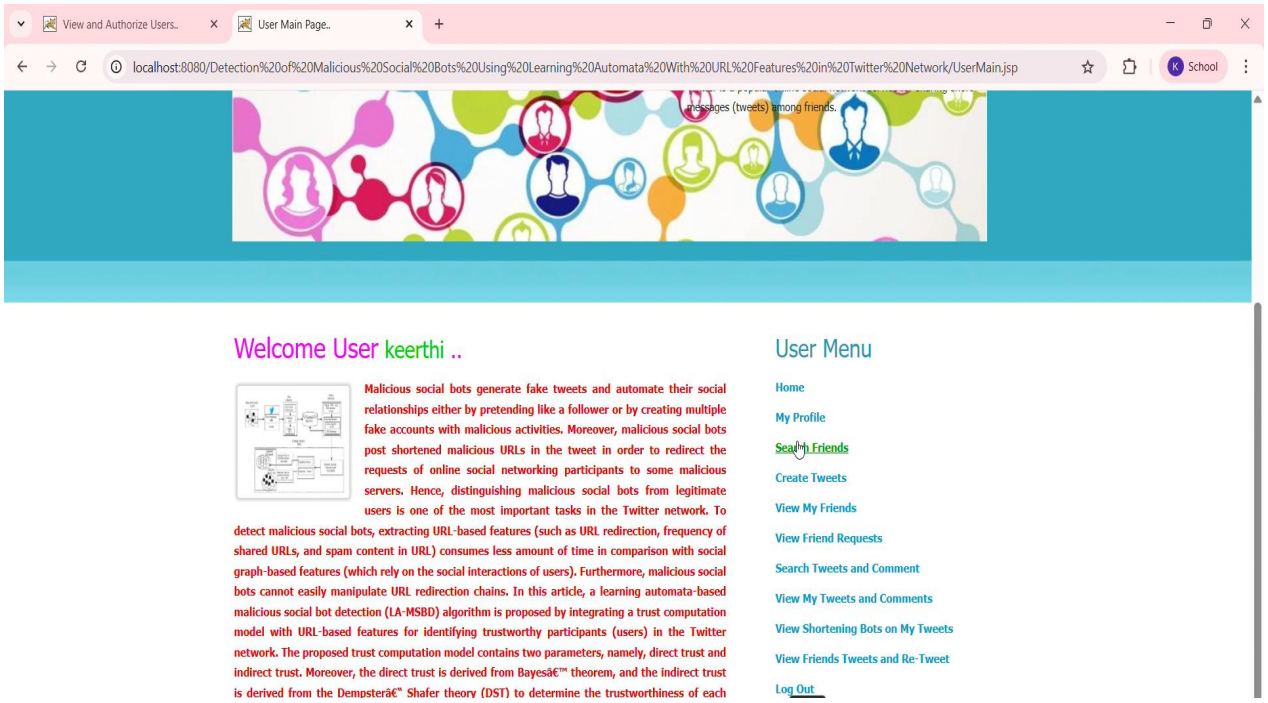
User Registration

User Registration Page



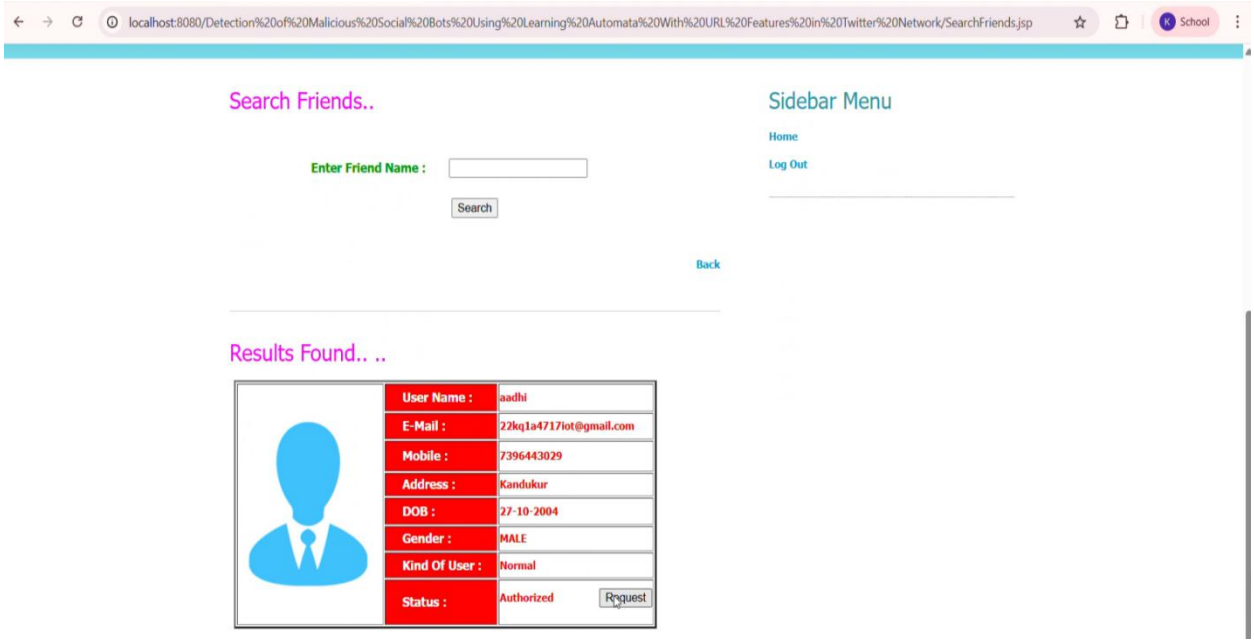
User Registrartion Status

User Menu Page



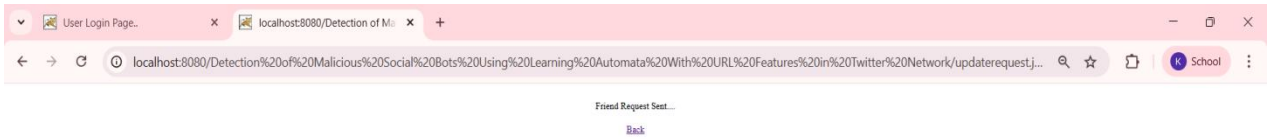
User Menu

Search Friends Page



Search Friends

Request Sent Page



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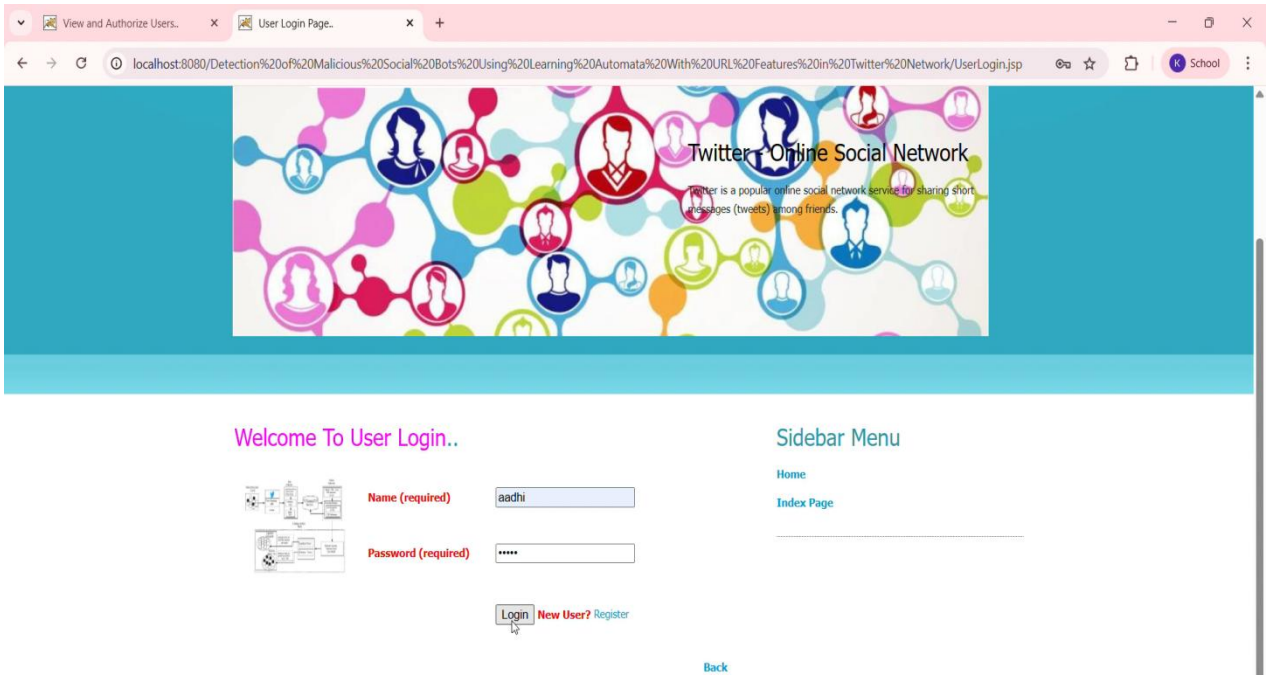
Uesr Login Page



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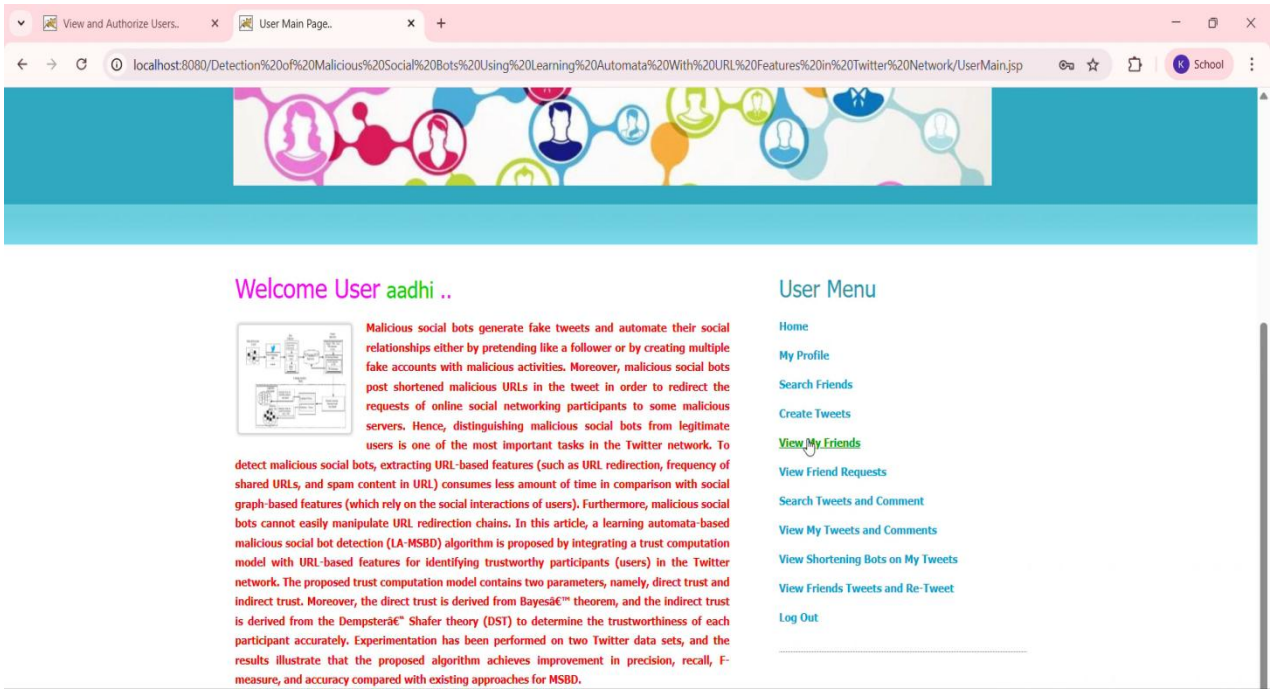
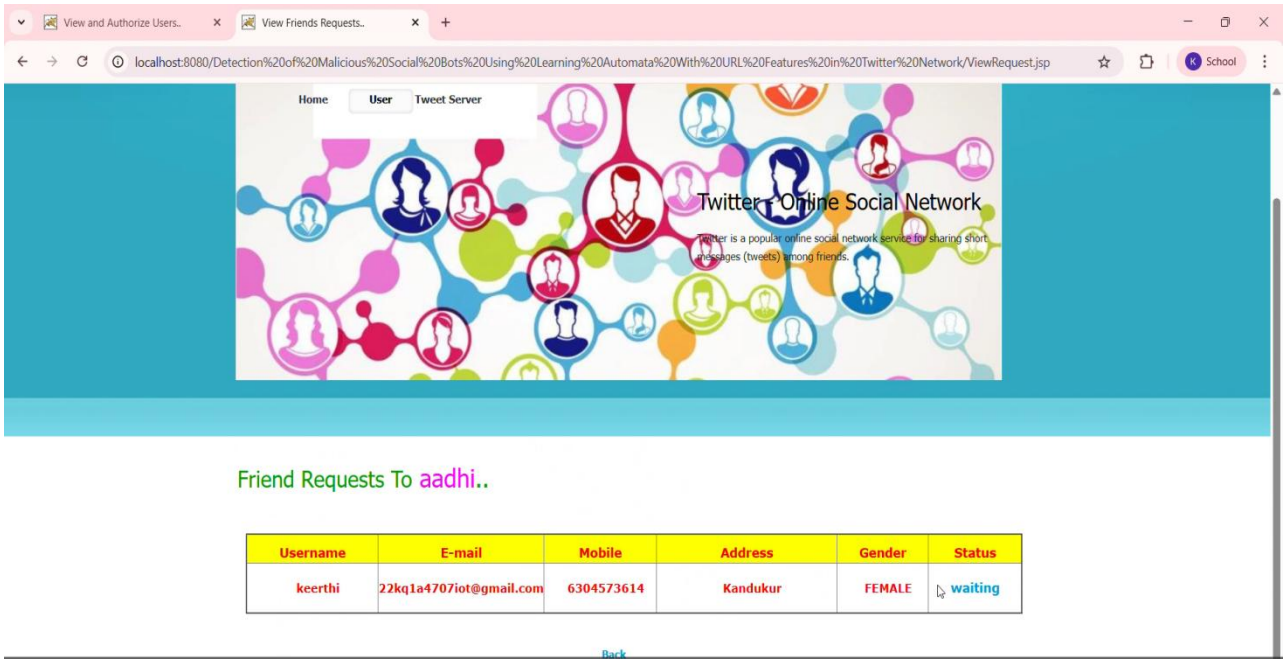
User Menu Page



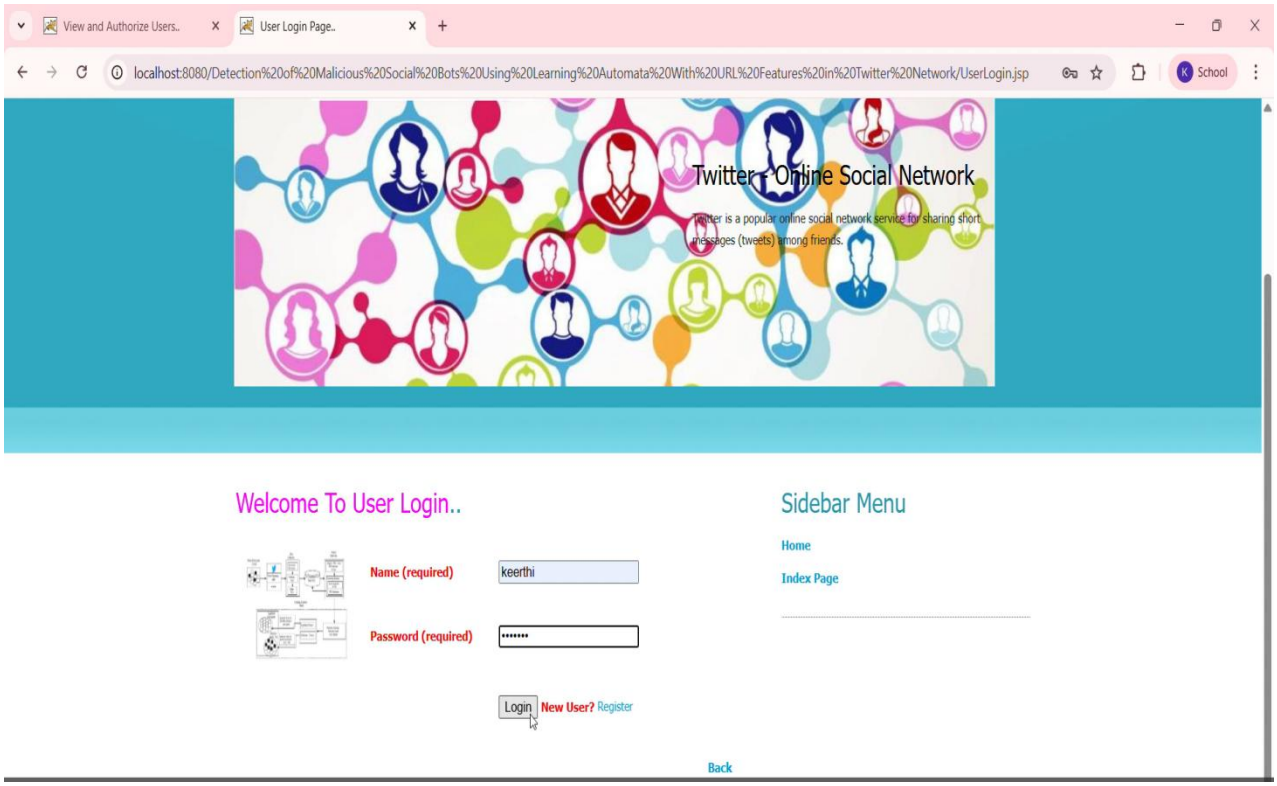
User Menu

Find Request



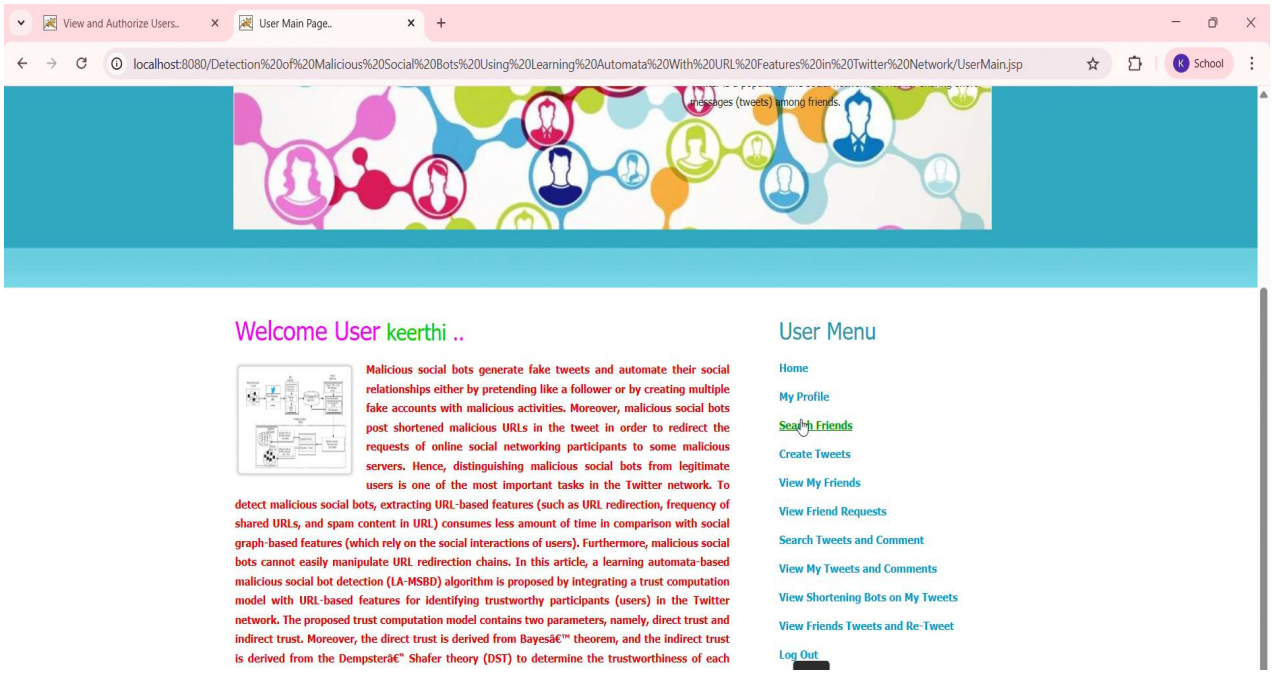
Find Request

Another User Login Page



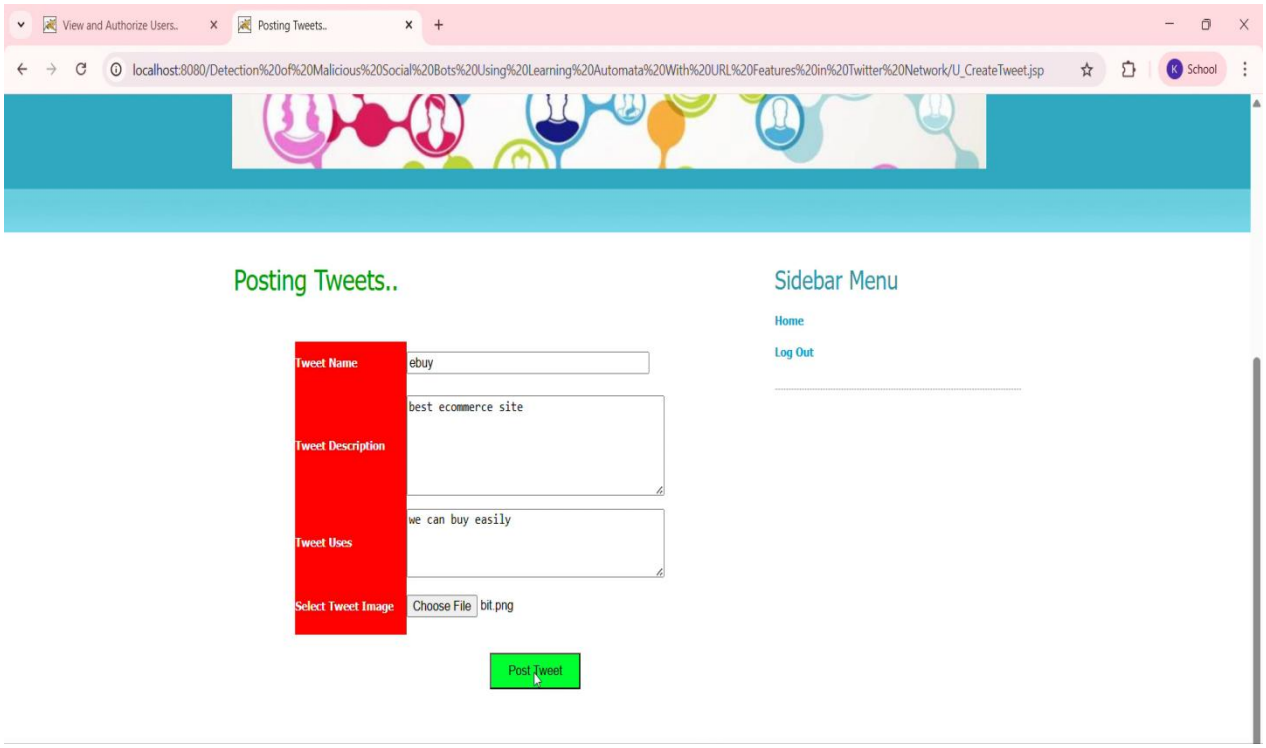
Another User Login

Another User Menu Page



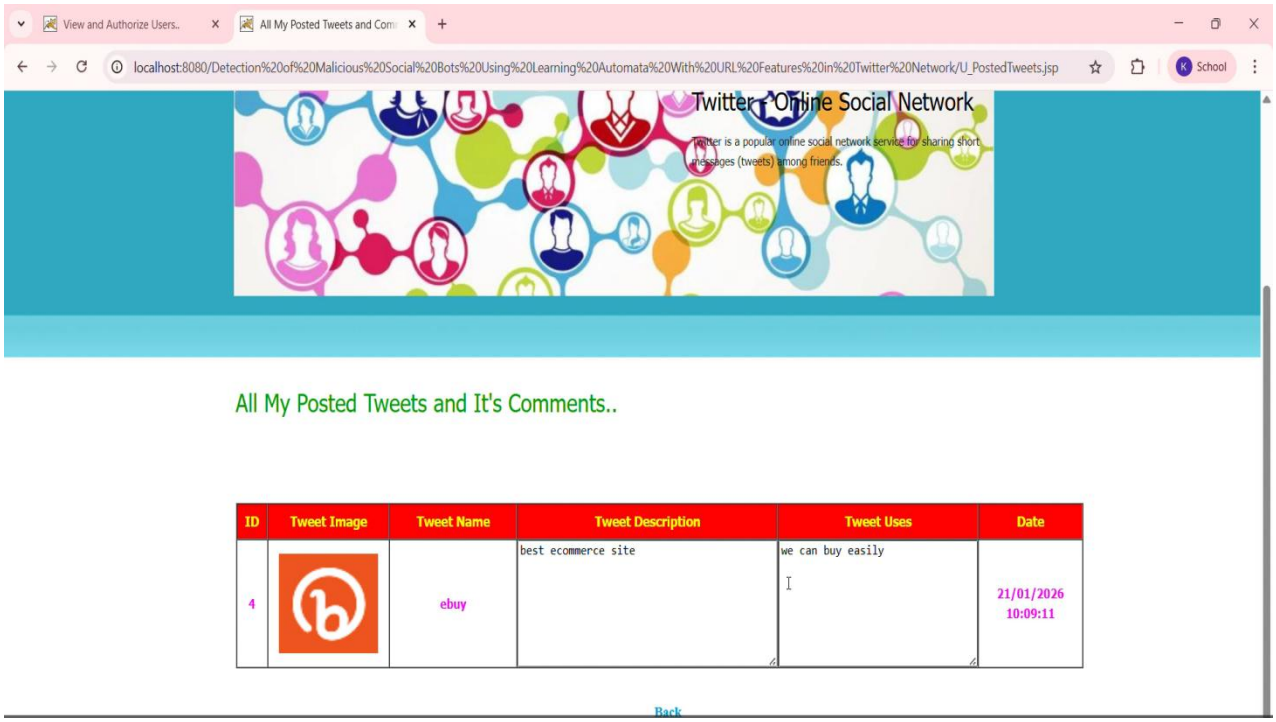
Another User Menu

Posting Tweets Page



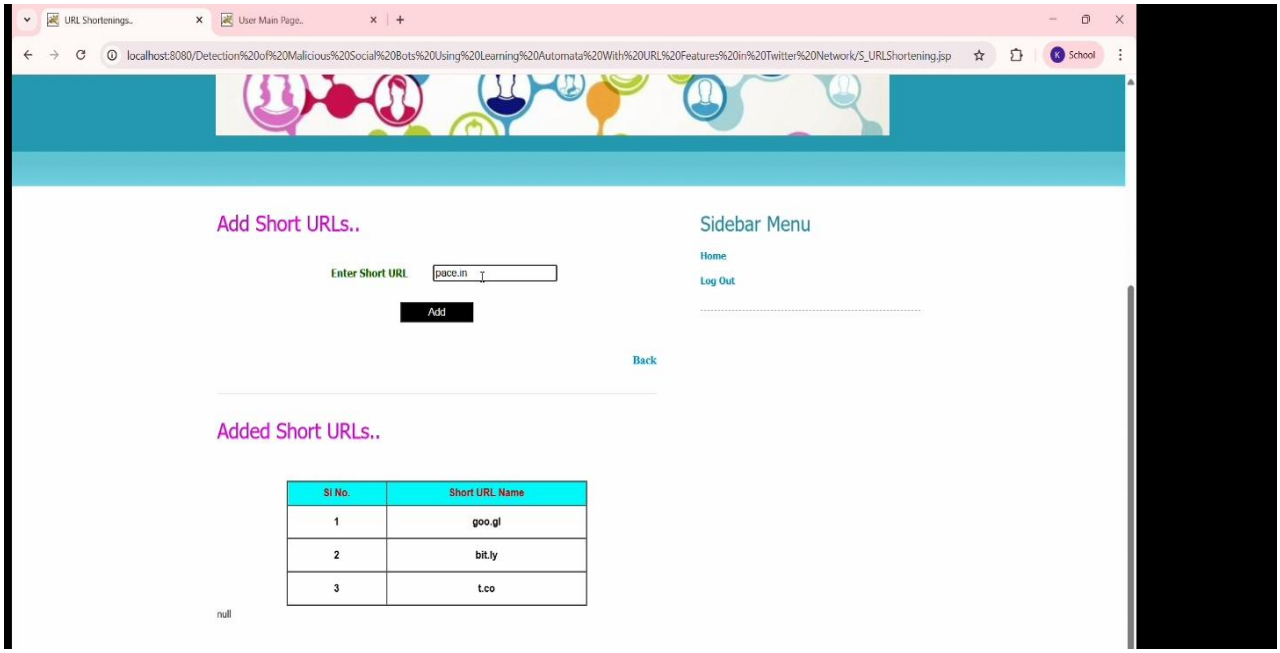
Posting Tweets

Posted Tweets & Comments Page



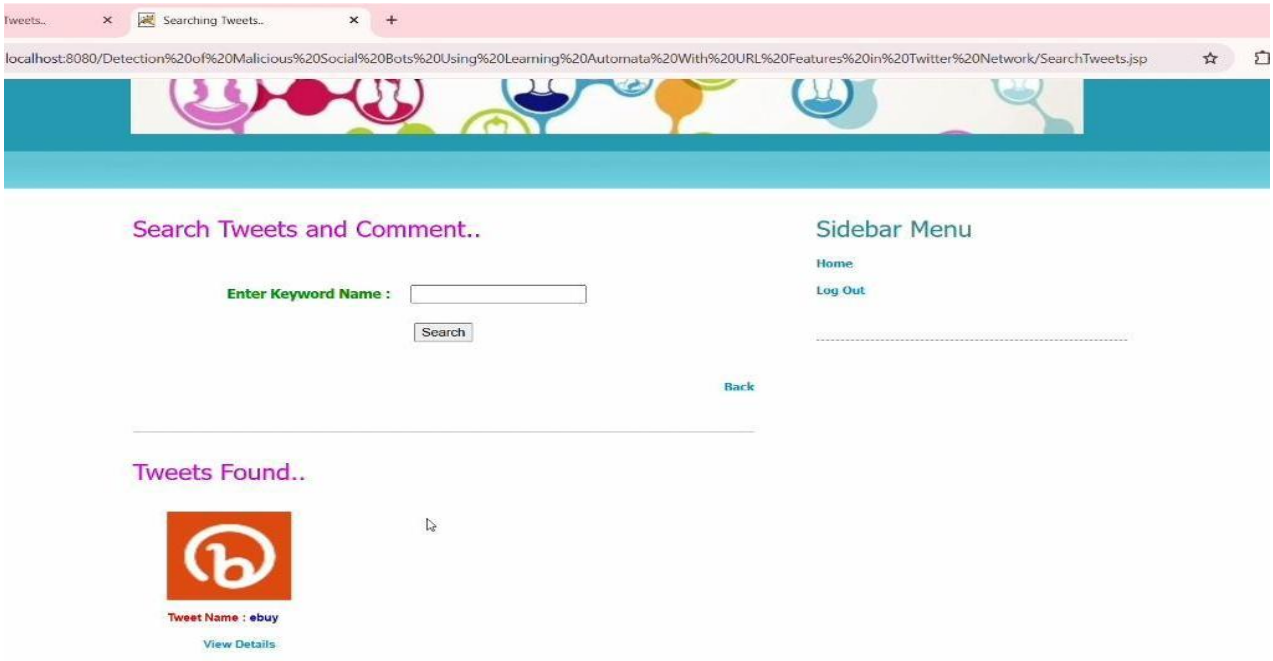
Posted Tweets & Comments

URLs Page



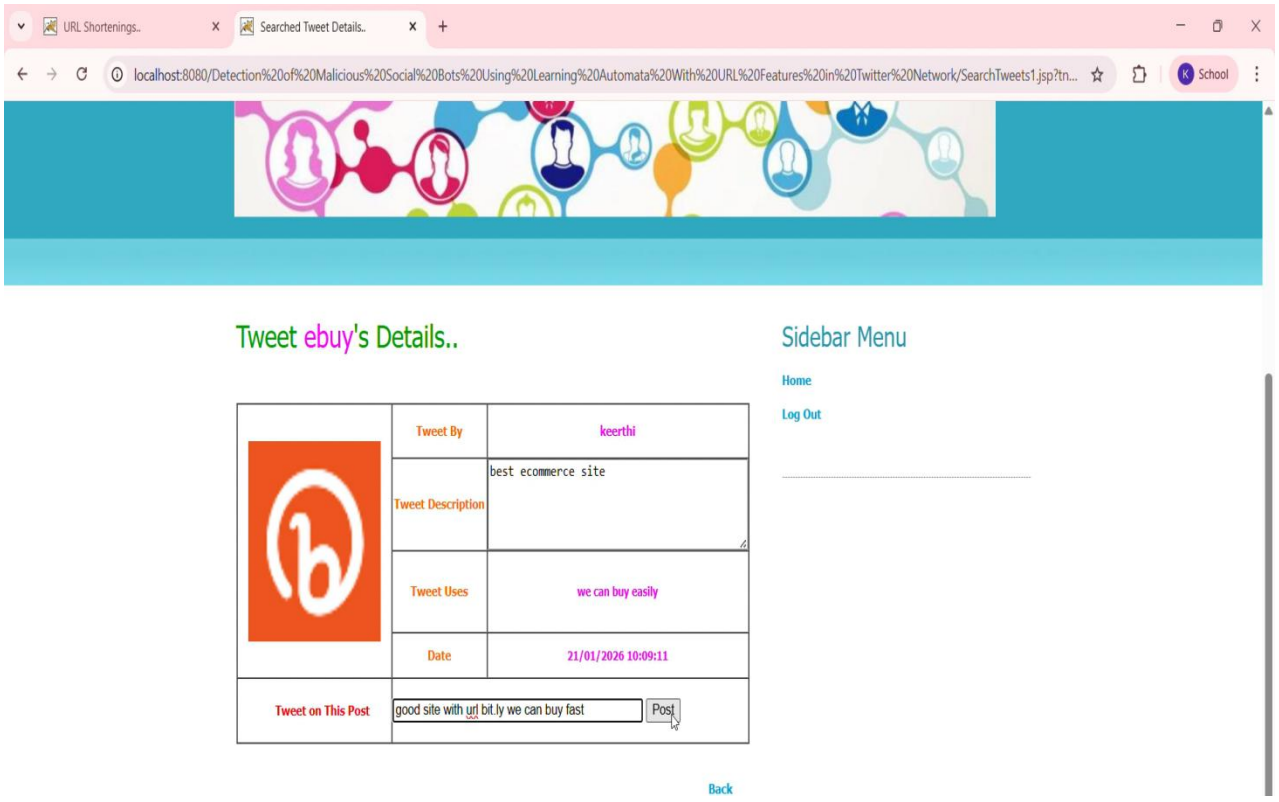
URLs

Search Tweets & Comments Page



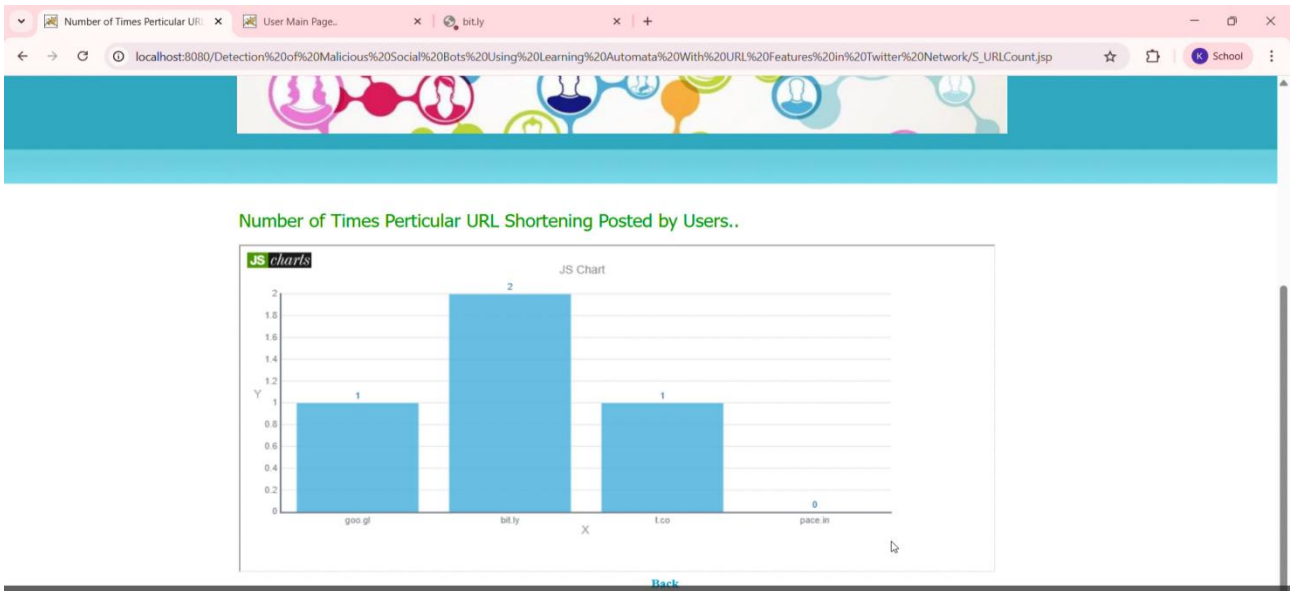
Search Tweets & Comments Page

Tweet Details Page



Tweet Details

No.of Times particular URL Posted by Users



No.of Times particular URL Posted by Users

8 . CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

In conclusion, the LA-MSBD algorithm provides a robust and adaptive framework for detecting malicious social bots by integrating reinforcement learning, trust computation, and URL feature analysis. By combining direct trust derived from Bayesian learning with indirect trust aggregated through Dempster–Shafer Theory, the model makes more reliable decisions even in the presence of noisy and uncertain social data. The incremental learning capability enables the system to adjust to evolving bot strategies instead of relying on static patterns.

The experimental results confirm that focusing on URL behavior and trust signals significantly strengthens detection performance compared to methods based only on profile or content features. The achieved improvements in accuracy, precision, and recall indicate that the approach is practical for real-world social network monitoring. The framework is also flexible enough to incorporate additional behavioral and network features in future extensions.

Overall, the proposed system contributes to improving security and reliability in online social networks by enabling early and accurate identification of malicious bot accounts. With further enhancements such as larger datasets, cross-platform analysis, and hybrid learning models, the approach can be extended into a scalable, real-time bot detection solution for next-generation social media environments.

8.2 Future Scope

The future scope of the **Malicious Bot Detection on Twitter Using Reinforcement Learning and URL Feature Analysis** project is very wide and can be extended in several directions. In addition to the existing scope, the project can be further extended by incorporating graph-based analysis to study the interaction networks among users. By analyzing follower–following relationships, retweet networks, and mention graphs, the system can identify coordinated bot campaigns and bot communities rather than detecting bots individually. This will help in uncovering large-scale malicious operations that work together to amplify misinformation or spam.

Another important future enhancement is the use of explainable AI (XAI) techniques. Providing clear explanations for why an account is classified as malicious or genuine will increase transparency and trust in the system, especially for analysts and decision-makers. Such explanations can also help security experts understand emerging attack patterns and refine detection strategies more effectively.

The system can also be adapted for multilingual environments by analyzing URLs and behavior across different languages and regions. This is particularly important as malicious bots often target global audiences. By supporting multiple languages and regional threat patterns, the detection framework can become more inclusive and globally applicable.

Finally, integrating the system with automated response mechanisms—such as flagging accounts, limiting reach, or notifying administrators—can make the solution more proactive. Instead of only detecting malicious bots, the system can help mitigate their impact in real time. These enhancements will make the project a comprehensive, intelligent, and future-ready solution for social media security.

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